The Dynamic Distribution in the Fixed Cost Model: An Analytical Solution*

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Abstract

I derive an analytical solution to the Kolmogorov forward equation for fixed cost models. In this PDE, the dynamic distribution depends on the endogenously determined flow of resetting agents. A shortcut determines this flow without first solving for the distribution, and aggregate variables can then be calculated from the flow directly. Steady-state conditional adjustment behavior recovers the entire marginal reset-flow response to a common state shift. In a fixed-cost investment application, aggregate dynamics require only a one-dimensional productivity-weighted distribution of capital gaps. The functional sufficient statistic reveals the marginal macroeconomic response to a corporate tax cut.

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1 Introduction

Many economic decisions require paying fixed costs. The microeconomics of these decisions is well understood. But the macroeconomics of fixed costs is challenging to study, because doing so often requires keeping track of an infinite-dimensional state variable: the dynamic distribution of agents. Its evolution is governed by a partial differential equation (PDE), the *Kolmogorov Forward Equation* (KFE).The solution is difficult to characterize analytically because it depends on the flow of resetting agents, which itself is determined endogenously from the dynamic distribution. Because of this endogenous feedback, theorists rely on approximations to characterize aggregate behavior.

To address these challenges, I derive an analytical solution for the dynamic distribution in a canonical fixed cost model. The key insight is that the evolution of the infinite-dimensional distribution can be summarized by a one-dimensional object: the endogenous flow of resetting agents.[1] Crucially, this reset flow can be determined without first solving for the dynamic distribution. This circumvents the challenging endogenous feedback. Once the flow is known, the dynamic distribution follows from a standard PDE solution. The reset-flow representation is especially useful in settings where endogenous "reinjection" cannot be ignored, such as models featuring asymmetry, drift, or aggregates that depend on even moments.[2]

The reset flow provides an especially valuable shortcut because paths of aggregate variables are linear in the path of the reset flow. This transformation depends only on the initial condition, model parameters, and function mapping the distribution to the aggregate. Therefore the full distribution is not necessary to compute in order to describe macroeconomic dynamics.

The analytical solution reveals another insight: steady-state adjustment behavior pins down the entire marginal impulse response function (IRF) to an aggregate shock. I derive a *functional sufficient statistic* for the reset-flow response to a marginal aggre-

[1] Quantitative models often approximate the infinite-dimensional distribution, for example Midrigan (2011) and others use the Krusell and Smith (1998) method to reduce the dimensionality. Earlier work used modeling assumptions that eliminated the distribution as an aggregate state variable (Caplin and Spulber, 1987; Caplin and Leahy, 1997).

[2] Alvarez and Lippi (2022) previously made substantial progress by assuming there is no "reinjection", i.e., agents leave the distribution after resetting. This method is useful for calculating IRFs of odd moments in models with symmetric inaction regions and no drift (e.g., no trend inflation or productivity growth) because the KFE reduces to the classic heat equation.

gate shock that shifts agents' states without changing decision rules. The ingredients are two steady-state objects: the reset-time density after adjustment and the average slope of the conditional survival function. The marginal reset-flow IRF then determines the marginal IRF of other macroeconomic variables. This result builds on the sufficient-statistic literature following Alvarez et al. (2016), which relates the cumulative impulse response (CIR) to steady-state adjustment statistics. The functional sufficient statistic applies to small shocks; when shocks are large, the analytical solution can instead be applied to a calibrated model.This is important because numerical solutions to the nonlinear model demonstrate that conclusions regarding small shocks do not necessarily generalize (Golosov and Lucas Jr., 2007; Cavallo et al., 2024).

The canonical model has constant coefficients. I also consider a more general setting with state-dependent drift, diffusion, and random resets. The three main theorems still apply, even if the closed-form expressions do not. The main restriction is that post-shock decision rules and the functions governing state dynamics remain fixed over time. The distributional solution allows parameter changes that cause a one-off movement in the inaction boundaries, but not boundaries that continue to change during the transition.[3]

This type of fixed cost model, with aggregate shocks and a dynamic distribution governed by a KFE, applies to a wide variety of settings, including investment adjustment costs (Baley and Blanco, 2021), hiring and firing costs (Elsby and Michaels, 2019), information acquisition costs (Alvarez et al., 2018), wage renegotiation costs (Blanco and Drenik, 2023), and famously menu costs (Golosov and Lucas Jr., 2007). As an example, I demonstrate the analytical solution in a standard investment model with fixed costs of adjustment. The firms' (S, s) strategies are well known; all of the insights relate to aggregation. I consider a corporate tax cut and show how steady-state adjustment functions map into the marginal IRFs of output and investment.

The application delivers one more valuable finding: it is not necessary to solve for the joint distribution of capital and productivity to compute aggregate dynamics. I show that a one-dimensional productivity-weighted distribution of capital gaps follows the same form of KFE as the canonical model, and detrended aggregate capital and output are functions of this weighted distribution. Therefore the main theorems

[3]Some general equilibrium models have time-varying boundaries. For example, in menu cost models, a fixed inaction region is a good approximation in standard calibrations (Alvarez and Lippi, 2014). However in other calibrations, particularly with high trend inflation, boundary movement matters; Adams (2026) adapts this paper's method to account for these general equilibrium effects.

apply; a productivity-weighted reset flow determines the aggregate IRFs in general, and their marginal IRFs are characterized by the functional sufficient statistic.

Section 2 describes the canonical fixed cost model. Section 3 derives the reset-flow representation and solution (Theorem 1). Section 4 demonstrates how to express aggregate dynamics as linear transformations of the reset flow (Theorem 2). Section 5 derives the functional sufficient-statistic result giving the marginal IRF from steady-state adjustment statistics (Theorem 3). Section 6 applies the solution to a fixed-cost investment model. Section 7 concludes.

2 The Fixed Cost Model

This section introduces the canonical fixed cost model, abstracting from decision-making microfoundations to focus on aggregation and the dynamic distribution.[4]

I develop the analytical solution in three steps. This section defines the reset flow and Lemma 1 states the KFE conditional on that flow. Then Theorem 1 uses the Green's function to solve directly for the reset flow, after which the distribution can be recovered. Finally Theorem 2 shows that many aggregate dynamics can be calculated from the reset flow alone, without first recovering the distribution.

There is a continuum of agents i; at any time t, each agent is characterized by a state variable $x_i(t)$. Each agent's state variable follows a diffusion:

$$dx_i(t) = \mu dt + \sigma dW_i(t) \tag{1}$$

where $W_i(t)$ is Brownian motion that is independent across agents, and μ captures a common trend. For notational clarity, I drop the i and t arguments henceforth.

When an agent's state is sufficiently low ($x \leq 0$) or sufficiently high ($x \geq \bar{x}$), it is willing to pay a fixed cost and reset its state to $x = x^*$. The interval $[0, \bar{x}]$ with $0 < x^* < \bar{x}$ is assumed to be constant.[5] Agents also randomly reset at rate $\eta \geq 0$, leaving the distribution and reentering at $x = x^*$.

The distribution of agents' states is $h(x,t)$, a function on $x \in [0, \bar{x}]$ and $t \geq 0$. Solving the fixed cost model entails finding the function $h(x,t)$ that satisfies a number of conditions:

[4] See Stokey (2008) for a textbook treatment.

[5] Because x^* must be in the interior, this rules out models where agents exit one side of a region to reappear at the other, such as Caplin and Spulber (1987).

1. The distribution satisfies the KFE:[6]

$$\partial_t h(x,t) = \gamma \partial_x^2 h(x,t) - \mu \partial_x h(x,t) - \eta h(x,t) \tag{2}$$

on the interval $[0, x^*) \cup (x^*, \bar{x}]$, where $\gamma \equiv \frac{\sigma^2}{2}$.

2. The *continuity condition*: while $h(x,t)$ might not be differentiable in x at x^*, it must be continuous for $t > 0$.

3. The bounds 0 and $\bar{x}$ are absorbing barriers, implying the Dirichlet boundary conditions

$$h(0,t) = 0 \qquad h(\bar{x},t) = 0$$

4. The distribution is consistent with the initial condition

$$h(x,0) = \phi(x)$$

where $\phi(x)$ is a probability measure, possibly with point masses.

5. Probability is conserved, i.e., for all t,

$$\int_0^{\bar{x}} h(x,t)\,dx = 1$$

This structure has been used to study economic inaction in a wide variety of settings.

The model differs from a standard diffusion because agents who leave through either boundary or reset randomly immediately reenter at x^*. Define the *reset flow* $F(t)$ as

$$F(t) \equiv \gamma \partial_x h(0,t) - \gamma \partial_x h(\bar{x},t) + \eta \tag{3}$$

The first two components capture the probability flow across the boundaries, while η is the flow of random adjustments.[7] These agents reenter the distribution at x^*, creating a point-like source where probability enters at rate $F(t)$.

Lemma 1 formalizes how the reset flow definition follows from the conservation assumption $\int_0^{\bar{x}} h(x,t)\,dx = 1$, extends the KFE to the entire inaction region, and gives all the conditions as a standard PDE problem.

[6] See (Stokey, 2008, Section 3.3) for an introduction to the KFE.

[7] In this paper, derivatives at boundaries denote one-sided derivatives from the interior.

Lemma 1. *Conditional on the reset flow $F(t)$, the model is given by the following PDE problem for $x \in [0, \bar{x}]$ and $t \geq 0$:*

$$\partial_t h(x,t) = \gamma \partial_x^2 h(x,t) - \mu \partial_x h(x,t) - \eta h(x,t) + \delta(x - x^*) F(t)$$

$$h(0,t) = 0 \qquad h(\bar{x},t) = 0$$

$$h(x,0) = \phi(x)$$

where $\delta(x - x^*)$ is the Dirac delta.

Proof: Appendix A.1

2.1 Economic Environments

The state variable x typically represents a gap between a past choice and an ideal target. Gaps evolve stochastically, and when the gap gets large enough, agents are willing to pay a fixed cost and reset to x^*: an (S, s) model. This structure applies to a wide variety of settings.

A classic example is investment, for which inaction and large discrete adjustments are observed in the data. The model I explore in Section 6 is a special case of the investment model studied and estimated by Baley and Blanco (2021).

Sticky prices are another common example, where x represents the gap between a firm's current and desired price. Menu costs generate the inaction region; following Golosov and Lucas Jr. (2007), Nakamura and Steinsson (2010), and others, Alvarez and Lippi (2022) provide an example of the continuous-time formulation.

With fixed hiring and firing costs, x is an employment gap, and firms adjust employment only when it is sufficiently far from its target (Elsby and Michaels, 2019). Inventory management also has an (S, s) structure: x is an inventory of finished products, which declines stochastically until a fixed ordering cost is paid and the stock is replenished (Fisher and Hornstein, 2000). In durable replacement decisions, x can measure the age or condition of a machine that is costly to fix or replace (Cooper et al., 1999).

Rational inattention is another friction that can create inaction. When an agent's state is close to its target, the gain from acquiring information and reconsidering its choice is small; as the gap grows, more frequent review becomes valuable. Woodford (2009) applies this idea to state-dependent price-setting, and Alvarez et al. (2022)

show that information-acquisition costs imply that the KFE includes a generalized adjustment hazard. This does not necessarily fit in the baseline framework, where the random reset rate is a constant η. However, some results still generalize, as I discuss next.

2.2 Generalization and Caveats

How general is this fixed cost model? The constant coefficients in equation (2) are a common case. But what if the coefficients are allowed to be state-dependent? Assume $\gamma(x) > 0$ and $\eta(x) \geq 0$, with the coefficients sufficiently regular. Then the more general KFE becomes

$$\partial_t h(x,t) = \partial_x^2 \left(\gamma(x)h(x,t)\right) - \partial_x \left(\mu(x)h(x,t)\right) - \eta(x)h(x,t) + \delta(x - x^*)F(t) \tag{4}$$

while the mass-preserving flow becomes

$$F(t) = \partial_x \left(\gamma(x)h(x,t)\right)\big|_{x=0} - \partial_x \left(\gamma(x)h(x,t)\right)\big|_{x=\bar{x}} + \int_0^{\bar{x}} \eta(x)h(x,t)\,dx \tag{5}$$

Without constant coefficients, the exact closed-form expressions I derive no longer apply. But the main practical insights still do: it is sufficient to find the reset flow $F(t)$ in order to characterize the solution. Theorems 1, 2, and 3 still hold without invoking the constant-coefficient expressions.

A real caveat is that the model assumes constant inaction boundaries. This is a feature of many models such as the fixed-cost investment example in Section 6. In other cases, it is merely a simplification.[8] The current approach *can* be used to address shocks that have discrete permanent effects on the boundaries; Supplemental Appendix F is an example. But models with truly dynamic boundaries require additional work. To address this, Adams (2026) builds on insights in the present paper to solve the full mean field game, including optimal dynamic boundaries.

[8]Particularly in menu cost models, it is common to assume that shocks do not affect the inaction region and that firms follow steady-state decision rules. Alvarez and Lippi (2014) and Cavallo et al. (2024) show that this is an accurate approximation for standard parameters. Moreover, in the general KFE, the state-dependent functions must be fixed over time. For example, this applies to the hazard functions studied by Alvarez et al. (2022), but they acknowledge that in a rational inattention model of sticky prices the fixed hazard abstracts from general equilibrium effects.

3 Solution

In this section, Theorem 1 shows how the infinite-dimensional distribution can be summarized by the one-dimensional reset flow $F(t)$. I first use the Green's function to express the distribution conditional on the reset flow, then use that expression to solve for the flow.

3.1 The Green's Function

A Green's function is a distributional IRF. $G(x, y, t)$ gives the density over x at horizon t generated by a unit mass initially located at y, for agents who have not yet reset. This distribution evolves based only on the stochastic processes; because it excludes endogenous resets, it is a standard building block in math and physics.[9] Solutions for arbitrary initial densities and reset flows can then be constructed by varying the inputs y and t.

If the reset flow is known, Property 1 gives the full dynamic distribution:

Property 1. *Given $F(t)$, the Green's function solution to the KFE is*

$$h(x,t) = \int_0^{\bar{x}} \phi(y)\, G(x,y,t)\, dy + \int_0^t F(\tau)\, G(x, x^*, t-\tau)\, d\tau \tag{6}$$

The first term propagates the initial density. The second adds the distributions generated by cohorts that reset at each earlier time τ. This is the standard Green's function solution for a non-homogeneous linear PDE.[10]

3.2 The Reset Flow

Equation (6) gives the distribution conditional on $F(t)$, but there is a feedback: $F(t)$ itself is endogenously determined by the distribution. This section's central result is that the reset flow can instead be determined in isolation without first solving for $h(x,t)$.

To do so, I introduce some useful objects that will reappear in Section 5's functional sufficient statistic. The conditional survival function $S(t \mid y)$ is the probability

[9] Supplemental Appendix B gives the general form for $G(x, y, t)$ and its closed-form expression in the constant-coefficient case.

[10] See (Polyanin, 2001, Sec. 0.4.2–4, 0.7.1–2, and 3.1.5) for equation (34), (6), and the constant-coefficient form, respectively.

that an agent beginning at y has not reset by time t. It is related to the Green's function by

$$S(t \mid y) \equiv \int_0^{\bar{x}} G(x, y, t)\, dx \tag{7}$$

The corresponding conditional reset-time density is

$$m(t \mid y) \equiv -\partial_t S(t \mid y) \tag{8}$$

Define

$$A(t) \equiv \int_0^{\bar{x}} m(t \mid y)\phi(y)\, dy \qquad B(t) \equiv m(t \mid x^*) \tag{9}$$

The function $A(t)$ represents the density of time to the first resets out of the initial distribution, while $B(t)$ is the density of time until the next reset after an agent jumps to x^*.

In continuous time, it is often useful to work in the frequency domain rather than the time domain. The following theorem gives the reset flow and its *Laplace transform* $\hat{F}(s)$. The Laplace transform of a function is defined by

$$\hat{F}(s) \equiv \mathcal{L}\{F\}(s) = \int_0^{\infty} F(t)\, e^{-st}\, dt$$

Theorem 1 (Reset Flow)**.** *The reset flow satisfies*

$$F(t) = A(t) + \int_0^t B(t - \tau)F(\tau)\, d\tau \tag{10}$$

for $t > 0$, with Laplace transform

$$\hat{F}(s) = \frac{\hat{A}(s)}{1 - \hat{B}(s)} \tag{11}$$

Proof: Appendix A.2

The first term in equation (10) captures agents' first adjustments after $t = 0$. The second accounts for subsequent adjustments: agents who reset at τ return to x^* and reset again after some time with density $B(t - \tau)$. In the frequency domain, convolution becomes multiplication, so the reset flow satisfies $\hat{F}(s) = \hat{A}(s) + \hat{B}(s)\,\hat{F}(s)$. Theorem 1 solves this scalar equation directly, without using the entire distribution $h(x, t)$.

Theorem 1 applies to Section 2.2's generalized model. For the constant-coefficient baseline, Supplemental Appendix B gives closed-form expressions for $\hat{A}(s)$ and $\hat{B}(s)$.

Once the reset flow is known, the distribution follows from Property 1. Corollary 1 states the solution, where $\hat{G}(x, y, s)$ is the Laplace transform of the Green's function.

Corollary 1. *The unique dynamic distribution is given by equation* (6)*, with* $F(t)$ *determined by Theorem 1. Thus, its Laplace transform is*

$$\hat{h}(x, s) = \int_0^{\bar{x}} \phi(y)\, \hat{G}(x, y, s)\, dy + \hat{G}(x, x^*, s)\, \frac{\hat{A}(s)}{1 - \hat{B}(s)} \tag{12}$$

Proof: Appendix A.3

Supplemental Appendix B.1 contains the standard expressions for $\hat{G}(x, y, s)$; then the inverse Laplace transform returns the dynamic distribution $h(x, t)$.

Still, the next section shows that the distribution need not be computed at all for many aggregate outcomes.

4 Using the Flow Function to Calculate Aggregate Dynamics

Many aggregate dynamics can be calculated from the reset flow without constructing the full distribution. Theorem 2 shows that this shortcut is a linear relationship mapping the reset flow path to the aggregate path.

Consider an aggregate variable $Z(t)$ (or some transformation thereof) that is determined from the distribution $h(x, t)$ by

$$Z(t) = \int_0^{\bar{x}} f_Z(x)\, h(x, t)\, dx \tag{13}$$

for some function $f_Z(x)$. Section 4.2 will discuss some common examples.[11]

Equation (6) implies that $Z(t)$ can be calculated from $F(t)$ without first constructing $h(x, t)$.

[11] The representation in equation (13) also reveals the relationship with Alvarez and Lippi (2022). When $\mu = 0$, $\bar{x} = 2x^*$, and $f_Z(x)$ is odd around x^*, the IRF can be found easily by solving the model without "reinjection". Without reinjection, the reset flow is $F(t) = 0$, and the PDE reduces to a standard homogeneous heat equation.

4.1 Frequency-Domain Macroeconomic Aggregates

The shortcut is cleanest in the frequency domain: Theorem 2 shows that $\hat{Z}(s)$ is *linear* in $\hat{F}(s)$, while Corollary 2 gives the equivalent time-domain representation.

Theorem 2 (Linearity)**.** *The Laplace transform $\hat{Z}(s)$ of an aggregate variable $Z(t)$ given by equation* (13) *is linear in the Laplace transform of the reset flow $\hat{F}(s)$:*

$$\hat{Z}(s) = \hat{A}^Z(s) + \hat{B}^Z(s)\,\hat{F}(s)$$

where $\hat{A}^Z$ and $\hat{B}^Z$ are the Laplace transforms of

$$\begin{aligned} A^Z(t) &\equiv \int_0^{\bar{x}} \phi(y) \left(\int_0^{\bar{x}} f_Z(x) G(x, y, t)\, dx \right) dy \\ B^Z(t) &\equiv \int_0^{\bar{x}} f_Z(x) G(x, x^*, t)\, dx \end{aligned} \tag{14}$$

Proof: Appendix A.4

Beyond the flow $F(t)$ there are two other ingredients, both of which are known *ex ante*. The $\hat{A}^Z(s)$ term captures how the initial distribution affects the aggregate variable, from agents who have not yet reset. The $\hat{B}^Z(s)$ term captures the effect on the aggregate of agents resetting at x^*. These functions are calculated from the Green's function and aggregating function. Combining these objects gives the aggregate path $Z(t)$ without constructing the entire $h(x,t)$ function, as the corollary makes clear:

Corollary 2 (Time-Domain Aggregation)**.** *The aggregate variable $Z(t)$ is*

$$Z(t) = A^Z(t) + \int_0^t F(\tau)\, B^Z(t-\tau)\, d\tau$$

Proof: Follows from Theorem 2, where $\hat{A}^Z(s)$ and $\hat{B}^Z(s)$ are Laplace transforms of $A^Z(t)$ and $B^Z(t)$.

From Corollary 2's time path for the aggregate variable $Z(t)$, it is straightforward to calculate typically reported objects such as the IRF and CIR:

$$IRF^Z(t) \equiv Z(t) - \overline{Z} \qquad CIR^Z \equiv \int_0^\infty \left[Z(t) - \overline{Z} \right] dt$$

where $\overline{Z}$ is the post-shock long-run value. These objects are simple to compute once

the Laplace transform $\hat{Z}(s)$ is known. Supplemental Appendix B.4 gives expressions for the IRF and CIR using the partial fraction expansion of $\hat{Z}(s)$.

Corollary 2 is immediately useful in settings where $Z(t)$ itself is relevant (e.g., investment models). In other cases, the aggregate variable of interest must first be transformed into a $Z(t)$ that satisfies equation (13) (e.g., the Golosov-Lucas output gap). The transformation must then be undone to calculate the IRF of the original variable.

4.2 Examples of Aggregating Functions

In this section, I discuss common examples of aggregating functions $f_Z(x)$. Supplemental Appendix B.3 derives their associated θ_n^Z series.

The Average Exponential State: Many aggregate variables $Z(t)$ depend on an average exponential function of the state:

$$Z(t) = \int_0^{\bar{x}} e^{\epsilon x} h(x,t)\, dx \tag{15}$$

for some ϵ. For example, in Golosov and Lucas Jr. (2007) x is the log markup gap, and evaluating this integral gives an output gap (raised to some power). In investment models, x is the log capital-productivity ratio, and evaluating this integral gives the average capital-productivity ratio. Section 6.2 shows that a productivity-weighted density can instead be used to recover aggregate capital.

The Average State: In some cases, aggregate variables $Z(t)$ depend on the average state ($f_Z(x) = x$):

$$Z(t) = \int_0^{\bar{x}} x h(x,t)\, dx \tag{16}$$

It is also common to use the average state as an approximation of a nonlinear function. For example, in the menu cost model studied by Alvarez et al. (2024), the aggregate output gap is calculated as proportional to the average markup gap.

The Squared State: It may be valuable to calculate the dynamics of higher-order moments, for example, to study the dynamics of misallocation. Many applications depend on the average squared state ($f_Z(x) = x^2$):

$$Z(t) = \int_0^{\bar{x}} x^2 h(x,t)\, dx \tag{17}$$

For example, in Baley and Blanco (2026) and Adams et al. (2025), the average squared state measures capital misallocation. In Cavallo et al. (2023), it captures welfare costs of inflation due to misallocation.

5 A Functional Sufficient Statistic

The analytical solution also reveals new properties of aggregate dynamics. In this section, Theorem 3 shows that steady-state adjustment functions determine the reset flow's entire marginal IRF following a common shock to agents' states that leaves decision rules unchanged. Corollary 3 then maps this response to the IRFs of other macroeconomic variables.

The sufficient-statistic literature uses steady-state adjustment statistics to characterize how small shocks to the distribution affect aggregates in fixed cost models. This strategy has focused on the CIR of the average state in idealized settings. For example, Alvarez et al. (2016) derive the CIR following a monetary shock in a constant-coefficient menu cost model with zero drift and a symmetric inaction region. Subsequent work allows for nonzero drift and asymmetric policies (Baley and Blanco, 2021), and for generalized state-dependent hazards (Alvarez et al., 2022).

Theorem 3 extends the prior approaches by deriving a *functional sufficient statistic*: the entire marginal IRF of the reset flow is determined by steady-state objects. This result is *functional* because its empirical inputs are functions. It is a *sufficient statistic* because it applies to Section 2.2's generalized model without needing to solve the model or identify its parameters.

The necessary empirical objects are derived from the conditional survival function $S(t \mid y)$ and reset-time density $m(t \mid y)$ defined in equations (7) and (8). Recall that $B(t) = m(t \mid x^*)$ is the reset-time density after adjustment. Define the steady-state average conditional survival slope by

$$\acute{\mathbf{S}}(t) \equiv \int_0^{\bar{x}} \partial_y S(t \mid y) \bar{h}(y)\, dy \tag{18}$$

The functions $B(t)$ and $\acute{\mathbf{S}}(t)$ are the key ingredients to Theorem 3, and are plausibly measurable in panel datasets of adjustments such as those used by Caballero et al. (1995), Baley and Blanco (2021), or Alvarez et al. (2024).[12]

[12]Estimation requires observations linking current states y to future adjustment times. Once

Starting from the steady state, suppose an unanticipated shock at $t = 0$ increases every agent's state by Δ, leaving dynamics and decision rules unchanged. Let $F(t; \Delta)$ denote the resulting reset flow and define its marginal IRF as

$$F^{\Delta}(t) \equiv \left.\frac{\partial F(t;\Delta)}{\partial \Delta}\right|_{\Delta=0}$$

Theorem 3 (Functional Sufficient Statistic)**.** *The reset flow's marginal IRF satisfies*

$$F^{\Delta}(t) = -\partial_t \acute{\mathbf{S}}(t) + \int_0^t B(t-\tau) F^{\Delta}(\tau)\, d\tau \tag{19}$$

for $t > 0$, with Laplace transform

$$\hat{F}^{\Delta}(s) = \frac{-s\hat{\acute{\mathbf{S}}}(s)}{1 - \hat{B}(s)} \tag{20}$$

Proof: Appendix A.5

The first term in equation (19) captures how the common shift changes the timing of the next reset among inaction spells already underway. The convolution term accounts for the flow from agents who reenter at x^*. Recovering the full IRF requires richer adjustment information – the functions $B(t)$ and $\acute{\mathbf{S}}(t)$ – than the few steady-state moments used for CIR sufficient statistics.

Theorem 2 linearly maps F^{Δ} into other aggregate responses using the functions A^Z and B^Z. These objects can also be recovered from steady-state micro data through the *generalized survival function* $S_Z(t \mid y)$:

$$S_Z(t \mid y) \equiv S(t \mid y)\, \mathbb{E}[f_Z(x(t); 0) \mid x(0) = y, t_{\text{reset}} > t] \tag{21}$$

$$= \int_0^{\bar{x}} f_Z(x; 0) G(x, y, t)\, dx \tag{22}$$

where t_{reset} denotes the time until the next adjustment. I allow the aggregating

$S(t \mid y)$ is estimated, differentiation and averaging over the observed steady-state distribution give $B(t)$ and $\acute{\mathbf{S}}(t)$. Econometric strategies to estimate $S(t \mid y)$ exist: in studies of unemployment duration, Chetty (2008) estimates survival curves for discrete groups, while Delgado et al. (2022) develop methods for estimating survival distributions and their average derivatives conditional on continuous variables. In ongoing work, Adams and Egan (2026) estimate the functional sufficient statistic for household money demand.

function $f_Z(x;\Delta)$ in equation (13) to depend on the shock Δ,[13] so that the aggregate variable of interest is

$$Z(t;\Delta) = \int_0^{\bar{x}} f_Z(x;\Delta) h(x,t;\Delta)\, dx$$

Thus $\mathbb{E}[f_Z(x(t);0) \mid x(0) = y, t_{\text{reset}} > t]$ is the average baseline outcome among agents that started at y and have not reset by t. Setting $f_Z(x;0) = 1$ reduces to equation (7)'s generic $S(t \mid y)$.

Corollary 3 (Aggregate Functional Sufficient Statistic)**.** *The aggregate's marginal IRF, $Z^\Delta(t) \equiv \partial_\Delta Z(t;\Delta)|_{\Delta=0}$, satisfies*

$$\begin{aligned} Z^\Delta(t) = &\int_0^{\bar{x}} \partial_y S_Z(t \mid y) \bar{h}(y)\, dy + \int_0^t S_Z(t-\tau \mid x^*) F^\Delta(\tau)\, d\tau \\ &+ \int_0^{\bar{x}} \partial_\Delta f_Z(x;\Delta)|_{\Delta=0}\, \bar{h}(x)\, dx \end{aligned} \tag{23}$$

for $t > 0$, where F^Δ is given by Theorem 3.

Proof: Appendix A.6

Again, a panel dataset containing adjustments and states can be used to estimate F^Δ, S_Z, and thus Z^Δ for any aggregating function of interest.

6 Example: Corporate Taxes and Investment

This section applies the analytical solution to a classic example: capital investment with fixed costs of adjustment. I analyze a partial-equilibrium model reminiscent of Caballero and Engel (1999); firms take interest rates as given, as in a small sector or small open economy. I consider a permanent cut to the corporate tax rate and use it to show how steady-state adjustment functions determine the marginal IRFs of macroeconomic aggregates.

One valuable new finding is that characterizing aggregate dynamics does not require the joint distribution of capital and productivity. Instead, a one-dimensional, productivity-weighted distribution of capital gaps determines detrended capital and output. And because it follows the same form of KFE as the general model, all the main theorems apply.

[13] Assume that $f_Z(x;\Delta)$ is bounded with a bounded derivative in Δ.

6.1 Environment

The environment is a special case of the investment model in Baley and Blanco (2021). A continuum of competitive firms use capital K and productivity P to produce output Y by

$$Y(P,K) = P^{1-\alpha}K^{\alpha}, \qquad \alpha \in (0,1)$$

Firms pay a corporate tax rate $\tau \in [0,1)$ on their income.[14] Log productivity $p \equiv \log P$ follows

$$dp = g_p dt + \sigma_p dW$$

where W is a Wiener process that is independent across firms.

Firms must pay a fixed cost $\kappa q K^{\text{tar}}$ to adjust capital, which they can buy or sell at price q. The fixed cost scales with the frictionless target capital, which equates the after-tax marginal product to the user cost:

$$K^{\text{tar}} \equiv \left(\frac{(1-\tau)\alpha}{q(r+\zeta)}\right)^{1/(1-\alpha)} P \tag{24}$$

where r denotes the discount rate and ζ the depreciation rate.

The firm's problem is homogeneous of degree one, so the state variable reduces to a single dimension, log "normalized capital":

$$k \equiv \log(K/P) - \frac{\log(1-\tau)}{1-\alpha}$$

Firms follow an (S,s) policy with inaction region $[\underline{k}, \bar{k}]$ and reset point k^*. With this normalization, the firm's optimization problem is independent of the tax rate τ. Its Hamilton-Jacobi-Bellman equation and boundary conditions are standard.[15]

To map the model to Section 2's state space, define the *capital gap* x as

$$x \equiv k - \underline{k} \qquad \bar{x} \equiv \bar{k} - \underline{k} \qquad x^* \equiv k^* - \underline{k}$$

so that x resets to x^*, and its law of motion between resets is

$$dx = \mu\, dt - \sigma_p\, dW \qquad \mu \equiv -(\zeta + g_p)$$

[14] For simplicity, the tax base excludes depreciation, investment, and adjustment costs.
[15] For complete details, see Supplemental Appendix D

Inside the inaction region $[0, \bar{x}]$, the distribution of firms follows the KFE (2) with $\gamma = \frac{\sigma_p^2}{2}$ and $\eta = 0$.[16]

I measure time in years and calibrate parameters to match the capital gap adjustment statistics measured by Baley and Blanco (2021) for Chilean manufacturing firms. I choose typical values, setting $\alpha = 0.85$, normalizing $q = 1$, and letting $\zeta = 0.03$ to match Chilean depreciation. Rapid productivity growth requires a large discount rate $r = 0.16$ to keep firm value finite.[17] The initial tax rate $\tau_0 = 0.26$ reflects the mean pre-TCJA marginal effective tax rate reported by Chodorow-Reich et al. (2024). The remaining parameters match the average time between adjustments (2.510), average capital gap adjustment (0.239), and adjustment variance (0.069).[18] This implies $g_p = 0.065$, $\sigma_p = 0.190$, and a normalized fixed cost $\tilde{\kappa} = 14$.

6.2 Aggregation

The density $h(x,t)$ weights each firm equally. Aggregating capital and output instead requires weighting firms by their productivity. Let $f(p,x,t)$ denote the joint density of log productivity and capital gaps, and define aggregate productivity and the productivity-weighted gap density by

$$\begin{aligned} \bar{P}(t) &\equiv \int_0^{\bar{x}} \int_{-\infty}^{\infty} e^p f(p,x,t)\, dp\, dx \\ h_P(x,t) &\equiv \frac{\int_{-\infty}^{\infty} e^p f(p,x,t)\, dp}{\bar{P}(t)} \end{aligned} \tag{25}$$

and thus $\int_0^{\bar{x}} h_P(x,t)\, dx = 1$.

Let $K(t)$ and $Y(t)$ denote aggregate capital and output. I use a P subscript to denote their values relative to aggregate productivity:

$$\begin{aligned} K_P(t) &\equiv \frac{K(t)}{\bar{P}(t)} = (1-\tau)^{1/(1-\alpha)} \int_0^{\bar{x}} e^{\underline{k}+x} h_P(x,t)\, dx \\ Y_P(t) &\equiv \frac{Y(t)}{\bar{P}(t)} = (1-\tau)^{\alpha/(1-\alpha)} \int_0^{\bar{x}} e^{\alpha(\underline{k}+x)} h_P(x,t)\, dx \end{aligned} \tag{26}$$

[16] This differs from Baley and Blanco, who define the capital gap relative to the steady-state mean rather than $\underline{k}$. I set $\eta = 0$ to minimize the error in the covariance between adjustments and inaction duration. Supplemental Appendix H varies this parameter to allow for free adjustments.

[17] This is close to the average US corporate discount rate measured by Gormsen and Huber (2025).

[18] Supplemental Appendix C derives closed-form expressions for these objects.

These *detrended* aggregates have the exponential form (15). Since $\partial_t K = I - \zeta K$ and $\partial_t \bar{P} = g_P \bar{P}$, where $g_P \equiv g_p + \sigma_p^2/2$, detrended investment satisfies

$$I_P(t) \equiv \frac{I(t)}{\bar{P}(t)} = \partial_t K_P(t) + (g_P + \zeta) K_P(t) \tag{27}$$

The following proposition states a surprising result: the weighted density $h_P(x,t)$ is also a case of the canonical fixed cost problem in Section 2. This is valuable because the joint density of productivity and capital is not needed to describe aggregate dynamics.

Proposition 1. *The productivity-weighted gap density satisfies*

$$\partial_t h_P(x,t) = \tfrac{\sigma_p^2}{2} \partial_x^2 h_P(x,t) - \mu_P \partial_x h_P(x,t) + \delta(x - x^*) F_P(t) \qquad \mu_P \equiv \mu - \sigma_p^2 \tag{28}$$

where

$$F_P(t) \equiv \tfrac{\sigma_p^2}{2} \left(\partial_x h_P(0,t) - \partial_x h_P(\bar{x},t) \right)$$

is the productivity-weighted boundary flow. The density is continuous at x^*, satisfies $h_P(0,t) = h_P(\bar{x},t) = 0$, and integrates to one.

Proof: Appendix A.7

The main theorems therefore apply to $h_P(x,t)$. And Theorem 2 implies a further shortcut: aggregate dynamics follow directly from the weighted reset flow $F_P(t)$. For example, the Laplace transform of aggregate capital $K_P(t)$ is

$$\hat{K}_P(s) = \hat{A}^{K_P}(s) + \hat{B}^{K_P}(s) \hat{F}_P(s) \tag{29}$$

where $A^{K_P}(t)$ and $B^{K_P}(t)$ are defined by (14), applied to the weighted density. These functions can hypothetically be recovered from panel data by estimating the generalized conditional survival function with productivity weights.

Furthermore, Theorem 3 and Corollary 3 can be applied to productivity-weighted adjustment histories in order to calculate marginal IRFs, as the tax experiment below illustrates.

6.3 Macroeconomic Effects of a Corporate Tax Cut

A corporate tax cut is a practical illustration of the type of shock to which Theorem 3 applies: it shifts the average state while leaving the inaction region unchanged. It also serves as an example where the relevant aggregating function f_Z depends on the aggregate shock.

The economy begins in the steady state with tax rate τ_0. At $t = 0$, an unanticipated permanent tax cut reduces the rate to $\tau_1 < \tau_0$, leaving productivity, r, and q unchanged. Before firms adjust, every capital gap changes by

$$\Delta = -\frac{1}{1-\alpha} \log \frac{1-\tau_1}{1-\tau_0} < 0 \tag{30}$$

Because the normalized firm problem is independent of the tax rate, the inaction region and reset point remain fixed.

In order to calculate the marginal IRF, applying Theorem 3 to h_P requires productivity-weighted versions of the conditional survival functions. The generalized function is

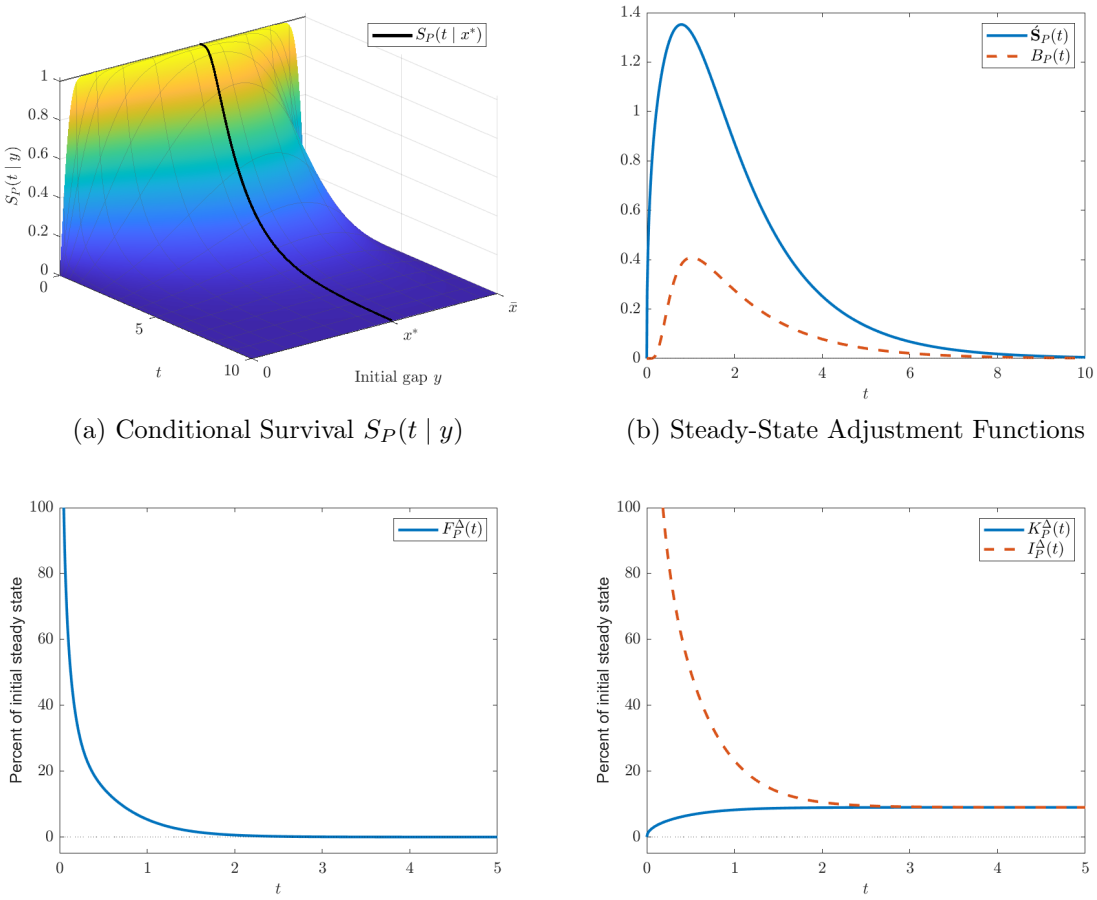
$$S_{Z,P}(t \mid y) = \int_0^{\bar{x}} f_Z(x; 0) G_P(x, y, t)\, dx$$

per equation (22), where $G_P(x, y, t)$ is the Green's function for Proposition 1's KFE. Setting $f_Z(x; 0) = 1$ gives $S_P(t \mid y)$, from which the functional sufficient statistic's inputs follow:

$$B_P(t) \equiv -\partial_t S_P(t \mid x^*) \qquad \acute{\mathbf{S}}_P(t) \equiv \int_0^{\bar{x}} \partial_y S_P(t \mid y) \bar{h}_P(y)\, dy$$

These functions can hypothetically be estimated from a panel of firms' capital, productivity, and adjustment dates, using productivity-weighted expectations in (21).

Thus steady-state adjustment functions determine the aggregate marginal response. Theorem 3 maps B_P and $\acute{\mathbf{S}}_P$ to the marginal reset-flow IRF $F_P^{\Delta}(t)$. Corollary 3 then gives $K_P^{\Delta}(t)$ and $Y_P^{\Delta}(t)$ using $S_{Z,P}$, $\bar{h}_P$, and the tax-dependent aggregating functions in (26). Per equation (30), the tax cut scales these functions by $e^{-\Delta}$ and $e^{-\alpha\Delta}$. Therefore the corollary's $\int_0^{\bar{x}} \partial_\Delta f_Z(x; \Delta)|_{\Delta=0} \bar{h}(x)\, dx$ term is $-\bar{K}_P$ for capital and $-\alpha\bar{Y}_P$ for output, where bars denote the initial steady state. And investment



(a) Conditional Survival $S_P(t \mid y)$

(b) Steady-State Adjustment Functions

(c) Reset Flow Marginal IRF

(d) Aggregate Marginal IRFs

Figure 1: Marginal Responses to the Tax Cut

Notes: Time is measured in years. Steady-state functions are productivity weighted. Marginal aggregate responses are scaled to a one-percentage-point tax cut.

follows from equation (27):

$$I_P^\Delta(t) = \partial_t K_P^\Delta(t) + (g_P + \zeta) K_P^\Delta(t) \tag{31}$$

These responses are per unit of Δ; multiplying by $-0.01/[(1-\alpha)(1-\tau_0)]$ expresses the marginal effect of a one-percentage-point tax cut.

Figure 1 demonstrates how steady-state adjustment behavior maps to marginal aggregate responses. Panel 1a presents the main ingredient: $S_P(t \mid y)$, with the dark line indicating the survival function conditional on a reset. The derivatives of this

surface are the functions in panel 1b. These slope functions are the key ingredients that determine the reset flow $F_P^{\Delta}(t)$ in panel 1c by Theorem 3. Then Corollary 3 maps this reset flow IRF to the aggregate IRFs in panel 1d. Investment spikes as the tax change causes firms to cross the inaction boundary at a high rate. Capital converges to its higher steady state as the distribution adjusts over time.

6.4 Additional Exercises

The analytical solution applies beyond the marginal level shocks covered by Theorem 3. Supplemental Appendix E illustrates how a larger tax cut changes the initial distribution and its subsequent evolution. Supplemental Appendix F explores instead a parameter change: a permanent increase in productivity growth. This modifies the inaction region and KFE, and generates a boom-lull investment pattern.

In Supplemental Appendix G I illustrate the importance of accounting for reinjection in the investment model. Ignoring reinjection substantially changes the reset flow, average gap, and aggregate responses to the tax cut.

Why is it valuable that Theorem 3 recovers the full IRF compared to a CIR? Many different response shapes can be consistent with the same cumulative value. Supplemental Appendix H demonstrates how, by varying the free adjustment rate in the tax cut exercise.

7 Conclusion

The endogenous reset flow is a useful focus point when studying aggregate dynamics in inaction models. It is one-dimensional, but summarizes the infinite-dimensional distribution's dynamics. And it reveals new connections between steady-state adjustment behavior and aggregate dynamics. This approach remains useful even when closed-form solutions are unavailable.

The functional sufficient statistic provides direction for future empirical work: estimating steady-state adjustment functions from micro data. Specifically, conditional survival functions determine the entire marginal response path of macroeconomic variables to aggregate shocks. Because this result holds in a general setting that applies to many inaction models, it implies that careful steady-state analysis will allow researchers to make macroeconomic predictions without first calibrating and solving

a structural model.

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A Proofs

Proofs are written in terms of Section 2.2's generalized model, except for Proposition 1, which is specific to the investment example.

A.1 Proof of Lemma 1

Proof. The boundary and initial conditions follow from the model. It remains to show that $F(t)$ is the intensity of the point source. The homogeneous KFE holds everywhere except at x^*. Its extension to the entire interval has the form

$$\partial_t h(x,t) = \partial_x^2 \left(\gamma(x)h(x,t)\right) - \partial_x \left(\mu(x)h(x,t)\right) - \eta(x)h(x,t) + \delta(x-x^*)R(t)$$

for some unknown function $R(t)$. The Dirac delta $\delta(x-x^*)$ indicates that the residual is only non-zero at x^*. Integrating over $[0,\bar{x}]$ and imposing probability conservation gives

$$\begin{aligned} 0 &= \partial_t \int_0^{\bar{x}} h(x,t)\,dx \\ &= \left[\partial_x \left(\gamma(x)h(x,t)\right) - \mu(x)h(x,t)\right]_0^{\bar{x}} - \int_0^{\bar{x}} \eta(x)h(x,t)\,dx + R(t) \end{aligned}$$

Using the absorbing-boundary conditions, equation (5) implies

$$R(t) = \partial_x \left(\gamma(x)h(x,t)\right)\big|_{x=0} - \partial_x \left(\gamma(x)h(x,t)\right)\big|_{x=\bar{x}} + \int_0^{\bar{x}} \eta(x)h(x,t)\,dx = F(t)$$

For constant coefficients, $\gamma(x) = \gamma$ and $\eta(x) = \eta$, so $\partial_x(\gamma(x)h(x,t)) = \gamma\partial_x h(x,t)$ and $\int_0^{\bar{x}} \eta(x)h(x,t)\,dx = \eta$, which gives the PDE and reset flow in the statement of the lemma. □

A.2 Proof of Theorem 1

Proof. Integrate equation (6) over x and substitute with equation (7) to get

$$\begin{aligned} 1 &= \int_0^{\bar{x}} h(x,t)\,dx \\ &= \int_0^{\bar{x}} S(t \mid y)\phi(y)\,dy + \int_0^t S(t-\tau \mid x^*)F(\tau)\,d\tau \end{aligned}$$

Differentiate with respect to t and use equations (8), (9), and $S(0 \mid x^*) = 1$ to yield

$$0 = -A(t) + F(t) - \int_0^t B(t-\tau)F(\tau)\,d\tau$$

Rearranging gives equation (10); taking the Laplace transform implies

$$\hat{F}(s) = \hat{A}(s) + \hat{B}(s)\hat{F}(s)$$

and solving this scalar equation gives $\hat{F}(s) = \frac{\hat{A}(s)}{1-\hat{B}(s)}$. □

A.3 Proof of Corollary 1

Proof. Taking the Laplace transform of equation (6) gives

$$\hat{h}(x,s) = \int_0^{\bar{x}} \phi(y)\,\hat{G}(x,y,s)\,dy + \hat{G}(x,x^*,s)\,\hat{F}(s)$$

Substituting equation (11) gives equation (12).

Equation (10) uniquely determines $F(t)$. Conditional on this flow, the KFE is a standard linear parabolic equation with a unique solution (Evans, 2022, Sec. 2.3 Thm. 5). □

A.4 Proof of Theorem 2

Proof. Substitute equation (6) into equation (13) and change the order of integration:

$$\begin{aligned} Z(t) &= \int_0^{\bar{x}} \phi(y) \left(\int_0^{\bar{x}} f_Z(x) G(x, y, t)\, dx \right) dy \\ &\quad + \int_0^t F(\tau) \left(\int_0^{\bar{x}} f_Z(x) G(x, x^*, t-\tau)\, dx \right) d\tau \\ &= A^Z(t) + \int_0^t F(\tau) B^Z(t-\tau)\, d\tau \end{aligned}$$

using the definitions of $A^Z(t)$ and $B^Z(t)$. The Laplace transform is

$$\hat{Z}(s) = \hat{A}^Z(s) + \hat{B}^Z(s)\hat{F}(s)$$

□

A.5 Proof of Theorem 3

Proof. Let $h(x, t; \Delta)$ denote the post-shock density and define its marginal response by

$$h^{\Delta}(x, t) \equiv \left. \frac{\partial h(x, t; \Delta)}{\partial \Delta} \right|_{\Delta=0}$$

$\bar{h}$ is zero outside $[0, \bar{x}]$. Following a positive shift Δ, the continuous component of the initial measure is $\bar{h}(x - \Delta)$, and the mass that crosses the upper boundary and resets to x^* is

$$M_{\Delta} = \int_{\bar{x}-\Delta}^{\bar{x}} \bar{h}(y)\, dy$$

The absorbing boundary imposes $\bar{h}(\bar{x}) = 0$, which implies $\frac{d}{d\Delta} M_{\Delta}\big|_{\Delta=0} = 0$. Thus to first order, the shock causes zero mass to reset, and

$$h^{\Delta}(x, 0) = -\partial_x \bar{h}(x) \tag{32}$$

The post-shock dynamics and policy are unchanged, so differentiating the KFE with respect to Δ at $\Delta = 0$ gives

$$\begin{aligned}\partial_t h^\Delta(x,t) = \partial_x^2\left(\gamma(x)h^\Delta(x,t)\right) - \partial_x\left(\mu(x)h^\Delta(x,t)\right) \\ - \eta(x)h^\Delta(x,t) + \delta(x-x^*)F^\Delta(t)\end{aligned}$$

and the Green's function is unchanged. Therefore by Property 1

$$h^\Delta(x,t) = -\int_0^{\bar{x}} G(x,y,t)\partial_y \bar{h}(y)\,dy + \int_0^t G(x,x^*,t-\tau)F^\Delta(\tau)\,d\tau$$

Integrating over x, integrating the first term by parts, and using $\bar{h}(0) = \bar{h}(\bar{x}) = 0$ gives

$$0 = \acute{\mathbf{S}}(t) + \int_0^t S(t-\tau \mid x^*)F^\Delta(\tau)\,d\tau$$

where the left-hand side is zero because the marginal perturbation integrates to zero.

Differentiate with respect to t and use $B(t) = -\partial_t S(t \mid x^*)$ and $S(0 \mid x^*) = 1$ to obtain equation (19). $\acute{\mathbf{S}}(0) = 0$, so its Laplace transform gives equation (20). □

A.6 Proof of Corollary 3

Proof. Since $h(x,t;0) = \bar{h}(x)$, differentiating the integral defining $Z(t;\Delta)$ gives

$$Z^\Delta(t) = \int_0^{\bar{x}} \partial_\Delta f_Z(x;\Delta)|_{\Delta=0}\,\bar{h}(x)\,dx + \int_0^{\bar{x}} f_Z(x;0)h^\Delta(x,t)\,dx$$

To evaluate the second term, hold the aggregating function fixed at $f_Z(x;0)$. Substituting the generalized survival function (22) into the definitions in equation (14) implies

$$\begin{aligned}A^Z(t) &= \int_0^{\bar{x}} S_Z(t \mid y)\phi(y)\,dy \\ B^Z(t) &= S_Z(t \mid x^*)\end{aligned}$$

The shock does not affect $B^Z(t)$, but does affect $A^Z(t)$ through the initial condition $h^\Delta(y,0) = -\partial_y \bar{h}(y)$, per equation (32). Integrate by parts to get

$$\partial_\Delta A^Z(t;\Delta)\big|_{\Delta=0} = -\int_0^{\bar{x}} S_Z(t \mid y)\partial_y \bar{h}(y)\, dy = \int_0^{\bar{x}} \partial_y S_Z(t \mid y)\bar{h}(y)\, dy$$

Differentiating the time-domain expression in Corollary 2 gives

$$\int_0^{\bar{x}} f_Z(x;0)h^\Delta(x,t)\, dx = \int_0^{\bar{x}} \partial_y S_Z(t \mid y)\bar{h}(y)\, dy + \int_0^t S_Z(t-\tau \mid x^*)F^\Delta(\tau)\, d\tau$$

Adding the direct term gives equation (23). □

A.7 Proof of Proposition 1

Proof. Away from the reset point, the KFE for the joint density is

$$\partial_t f = -g_p \partial_p f - \mu \partial_x f + \tfrac{\sigma_p^2}{2}\partial_p^2 f + \tfrac{\sigma_p^2}{2}\partial_x^2 f - \sigma_p^2 \partial_{px} f \tag{33}$$

Define $m_P(x,t) \equiv \int_{-\infty}^{\infty} e^p f(p,x,t)\, dp$, so that $h_P = m_P/\bar{P}$. For $x \neq x^*$, equation (33) and integration by parts give

$$\partial_t m_P(x,t) = (g_p + \tfrac{\sigma_p^2}{2})m_P(x,t) + \tfrac{\sigma_p^2}{2}\partial_x^2 m_P(x,t) - \mu_P \partial_x m_P(x,t)$$

In levels, P follows

$$dP = (g_p + \tfrac{\sigma_p^2}{2})P\, dt + \sigma_p P\, dW$$

Aggregate productivity grows exogenously by

$$\partial_t \bar{P}(t) = g_P \bar{P}(t)$$

where $g_P \equiv g_p + \sigma_p^2/2$. Differentiating $h_P = m_P/\bar{P}$ with respect to t implies

$$\partial_t h_P(x,t) = \tfrac{\sigma_p^2}{2}\partial_x^2 h_P(x,t) - \mu_P \partial_x h_P(x,t) \qquad x \neq x^*$$

Let $R_P(t)$ denote the unknown reset flow at x^*. Integrate the KFE over x:

$$0 = \tfrac{\sigma_p^2}{2}\left(\partial_x h_P(\bar{x},t) - \partial_x h_P(0,t)\right) + R_P(t) = -F_P(t) + R_P(t)$$

Thus $R_P(t) = F_P(t)$. Finally, the continuity and absorbing boundary conditions for f carry over to h_P. $\square$

Supplemental Appendix

B Formulas

This appendix gives eigenfunction expressions for the functions in Sections 3 and 4, followed by closed-form coefficients for the constant-coefficient model. These formulas make the analytical solution directly computable, but are not required for the general results.

B.1 General Eigenfunction Representation

In order to write the Green's function, first let $\mathcal{K}$ and $\mathcal{K}^\dagger$ denote the Kolmogorov operator and its adjoint. For a function $u(x)$, Section 2.2's generalized model gives

$$\begin{aligned}\mathcal{K}u &\equiv \partial_x^2\left(\gamma(x)u(x)\right) - \partial_x\left(\mu(x)u(x)\right) - \eta(x)u(x)\\ \mathcal{K}^\dagger u &\equiv \gamma(x)\partial_x^2 u(x) + \mu(x)\partial_x u(x) - \eta(x)u(x)\end{aligned}$$

Thus in the constant-coefficient model,

$$\mathcal{K} \equiv \gamma\partial_x^2 - \mu\partial_x - \eta \qquad \mathcal{K}^\dagger \equiv \gamma\partial_x^2 + \mu\partial_x - \eta$$

Let $\psi_n^r(x)$ and $\psi_n^\ell(x)$ denote the right and left eigenfunctions (for absorbing boundary conditions), with eigenvalue $-\lambda_n$, i.e.

$$\mathcal{K}\psi_n^r(x) = -\lambda_n\psi_n^r(x) \qquad \mathcal{K}^\dagger\psi_n^\ell(x) = -\lambda_n\psi_n^\ell(x)$$

Normalize the eigenfunctions so that

$$\int_0^{\bar{x}} \psi_n^\ell(x)\psi_m^r(x)\,dx = \mathbf{1}\{n = m\}$$

$n = 1, 2, \ldots$ indexes the eigenvalues of the Kolmogorov operator. The Green's function is the infinite sum

$$G(x, y, t) = \sum_{n=1}^{\infty} \psi_n^r(x)\,\psi_n^\ell(y)\,e^{-\lambda_n t} \tag{34}$$

Its Laplace transform is

$$\hat{G}(x, y, s) = \sum_{n=1}^{\infty} \frac{\psi_n^r(x)\, \psi_n^\ell(y)}{s + \lambda_n}$$

To evaluate the reset flow, define

$$\begin{aligned} a_n &\equiv \int_0^{\bar{x}} \psi_n^\ell(y)\, \phi(y)\, dy \\ b_n &\equiv \psi_n^\ell(x^*) \\ \theta_n &\equiv \gamma(0)(\psi_n^r)'(0) - \gamma(\bar{x})(\psi_n^r)'(\bar{x}) + \int_0^{\bar{x}} \eta(x)\psi_n^r(x)\, dx \end{aligned} \tag{35}$$

The coefficient θ_n gives the total reset flow generated by eigenmode n, accounting for both boundary and random resets. Integrating the eigenvalue equation over x and using the absorbing-boundary conditions gives

$$\theta_n = \lambda_n \int_0^{\bar{x}} \psi_n^r(x)\, dx$$

Thus, equations (7) and (8) imply

$$m(t \mid y) = \sum_{n=1}^{\infty} \theta_n \psi_n^\ell(y) e^{-\lambda_n t}$$

Substitute into equation (9) and take Laplace transforms to get the functions in Theorem 1:

$$\hat{A}(s) = \sum_{n=1}^{\infty} \frac{\theta_n\, a_n}{s + \lambda_n} \qquad \hat{B}(s) = \sum_{n=1}^{\infty} \frac{\theta_n\, b_n}{s + \lambda_n} \tag{36}$$

Substitute the eigenfunction expansion into equation (6) to yield

$$\begin{aligned} h(x, t) &= \sum_{n=1}^{\infty} \psi_n^r(x) c_n(t) \\ c_n(t) &\equiv a_n\, e^{-\lambda_n t} + b_n \int_0^t F(\tau)\, e^{-\lambda_n (t-\tau)}\, d\tau \end{aligned} \tag{37}$$

The reset flow is $F(t) = \sum_{n=1}^{\infty} \theta_n c_n(t)$. Taking the Laplace transform of the distribu-

tion and using equation (11) gives the equivalent expression for Corollary 1:

$$\begin{aligned}\hat{h}(x,s) &= \sum_{n=1}^{\infty} \psi_n^r(x)\, \frac{a_n + b_n\, \hat{F}(s)}{s+\lambda_n} \\ &= \sum_{n=1}^{\infty} \psi_n^r(x)\, \frac{\hat{A}(s)\, b_n + \left(1-\hat{B}(s)\right)\, a_n}{\left(1-\hat{B}(s)\right)(s+\lambda_n)}\end{aligned} \tag{38}$$

For an aggregating function f_Z (evaluated at $\Delta = 0$ when it depends on the shock), define

$$\theta_n^Z \equiv \int_0^{\bar{x}} f_Z(x)\, \psi_n^r(x)\, dx$$

The generalized survival function in equation (22) is then

$$S_Z(t \mid y) = \sum_{n=1}^{\infty} \theta_n^Z \psi_n^\ell(y) e^{-\lambda_n t}$$

Substitute equation (34) into equation (14) to yield the corresponding expressions for Theorem 2:

$$\hat{A}^Z(s) = \sum_{n=1}^{\infty} \frac{\theta_n^Z\, a_n}{s+\lambda_n} \qquad \hat{B}^Z(s) = \sum_{n=1}^{\infty} \frac{\theta_n^Z\, b_n}{s+\lambda_n}$$

B.2 The Constant-Coefficient Model

This section gives closed-form expressions for the constant-coefficient model.

Let $\chi \equiv \sqrt{2/\bar{x}}$ and $\nu \equiv \mu/(2\gamma)$. The eigenmodes are

$$\begin{aligned}\psi_n^r(x) &\equiv \chi e^{\nu x} \sin\left(\frac{n\pi x}{\bar{x}}\right) \\ \psi_n^\ell(x) &\equiv \chi e^{-\nu x} \sin\left(\frac{n\pi x}{\bar{x}}\right) \\ \lambda_n &\equiv \frac{\gamma n^2 \pi^2}{\bar{x}^2} + \frac{\mu^2}{4\gamma} + \eta \qquad T_n(t) \equiv e^{-\lambda_n t}\end{aligned} \tag{39}$$

Direct substitution gives

$$\mathcal{K}\psi_n^r(x) = -\lambda_n \psi_n^r(x) \qquad \mathcal{K}^\dagger \psi_n^\ell(x) = -\lambda_n \psi_n^\ell(x)$$

Moreover,

$$\int_0^{\bar{x}} \psi_n^{\ell}(x)\psi_m^{r}(x)\,dx = \frac{2}{\bar{x}}\int_0^{\bar{x}} \sin\left(\frac{n\pi x}{\bar{x}}\right)\sin\left(\frac{m\pi x}{\bar{x}}\right)\,dx = \mathbf{1}\{n=m\}$$

Thus the Green's function is

$$G(x,y,t) = \sum_{n=1}^{\infty} \psi_n^{r}(x)\psi_n^{\ell}(y)T_n(t)$$

The coefficients in equation (35) become

$$\begin{aligned}
a_n &= \chi\int_0^{\bar{x}} e^{-\nu y}\sin\left(\frac{n\pi y}{\bar{x}}\right)\phi(y)\,dy \\
b_n &= \chi e^{-\nu x^*}\sin\left(\frac{n\pi x^*}{\bar{x}}\right) \\
m_n &\equiv \int_0^{\bar{x}} \psi_n^{r}(x)\,dx = \frac{\chi(n\pi/\bar{x})\left(1-(-1)^n e^{\nu\bar{x}}\right)}{\nu^2+(n\pi/\bar{x})^2} \\
\theta_n^{\partial} &\equiv \gamma\left((\psi_n^{r})'(0) - (\psi_n^{r})'(\bar{x})\right) = \frac{\gamma\chi n\pi}{\bar{x}}\left(1-(-1)^n e^{\nu\bar{x}}\right) \\
\theta_n &= \theta_n^{\partial} + \eta m_n
\end{aligned} \tag{40}$$

The coefficient θ_n^{∂} gives resets at the boundaries, while ηm_n gives random resets. These coefficients can be substituted into equations (36)–(38) to calculate the reset flow and dynamic distribution.

B.3 Closed-Form Aggregating Coefficients

Proposition 2 gives the θ_n^Z sequence for each of the aggregating function examples described in Section 4.2. These series can be used with Theorem 2 to derive aggregate dynamics.

Proposition 2. *For a function $f_Z(x)$, the corresponding integrating sequence is*

$$\theta_n^Z = \int_0^{\bar{x}} f_Z(x)\sqrt{\frac{2}{\bar{x}}}e^{\frac{\mu}{2\gamma}x}\sin\left(\frac{\pi n x}{\bar{x}}\right)\,dx$$

Let $\omega_n = \nu^2 + \left(\frac{\pi n}{\bar{x}}\right)^2$. Then:

1. *For the exponential case* $f_Z(x) = e^{\epsilon x}$*:*

$$\theta_n^Z = \frac{\chi \pi n \left(1 - (-1)^n e^{(\epsilon+\nu)\bar{x}}\right)}{\bar{x}\left((\epsilon+\nu)^2 + (\frac{\pi n}{\bar{x}})^2\right)}$$

2. *For the linear case* $f_Z(x) = x$*:*

$$\theta_n^Z = \chi \frac{\pi n}{\omega_n}\Big[\frac{2\nu}{\bar{x}\omega_n}\Big((-1)^n e^{\nu\bar{x}} - 1\Big) - (-1)^n e^{\nu\bar{x}}\Big]$$

3. *For the quadratic case* $f_Z(x) = x^2$*:*

$$\theta_n^Z = \chi \frac{\pi n}{\bar{x}\omega_n^3}\Big[(-1)^n e^{\nu\bar{x}}\left(-\omega_n^2\bar{x}^2 + 4\nu\omega_n\bar{x} - 2\left(3\nu^2 - \left(\frac{\pi n}{\bar{x}}\right)^2\right)\right) + 2\left(3\nu^2 - \left(\frac{\pi n}{\bar{x}}\right)^2\right)\Big]$$

Proof. By the definition of $\psi_n^r(x)$, the desired sequences are given by

$$\theta_n^Z = \int_0^{\bar{x}} f_Z(x)\chi e^{\nu x} \sin\left(\frac{\pi n x}{\bar{x}}\right)\, dx$$

The following results are standard for exponential-sine integrals:

$$\int e^{cx}\sin(dx)\,dx = \frac{e^{cx}}{c^2+d^2}[c\sin(dx) - d\cos(dx)] \tag{41}$$

$$\int xe^{cx}\sin(dx)\,dx = \frac{e^{cx}}{(c^2+d^2)^2}\Big[\sin(dx)\Big(c(c^2+d^2)x - (c^2-d^2)\Big) + \cos(dx)\Big(-d(c^2+d^2)x + 2cd\Big)\Big] \tag{42}$$

$$\int x^2 e^{cx}\sin(dx)\,dx = \frac{e^{cx}}{(c^2+d^2)^3}\Big[\sin(dx)\Big(c(c^2+d^2)^2x^2 - 2(c^2-d^2)(c^2+d^2)x + 2c(c^2-3d^2)\Big) + \cos(dx)\Big(-d(c^2+d^2)^2x^2 + 4cd(c^2+d^2)x - 2d(3c^2-d^2)\Big)\Big] \tag{43}$$

In the exponential case, $f_Z(x) = e^{\epsilon x}$ implies

$$\theta_n^Z = \chi \int_0^{\bar{x}} e^{(\epsilon+\nu)x} \sin\left(\frac{\pi n x}{\bar{x}}\right) dx$$

Applying equation (41) with $c = \epsilon + \nu$ and $d = \frac{\pi n}{\bar{x}}$ and evaluating at the limits with $\sin(\pi n) = 0$ and $\cos(\pi n) = (-1)^n$ gives

$$\theta_n^Z = \chi \frac{\pi n}{\bar{x}\left((\epsilon+\nu)^2 + (\frac{\pi n}{\bar{x}})^2\right)} \left(1 - (-1)^n e^{(\epsilon+\nu)\bar{x}}\right)$$

In the linear case, $f_Z(x) = x$ implies

$$\theta_n^Z = \chi \int_0^{\bar{x}} x e^{\nu x} \sin\left(\frac{\pi n x}{\bar{x}}\right) dx$$

Apply equation (42) with $c = \nu$ and $d = \frac{\pi n}{\bar{x}}$, and evaluate at the limits:

$$\chi \int_0^{\bar{x}} x e^{\nu x} \sin\left(\frac{\pi n x}{\bar{x}}\right) dx = \chi \frac{\pi n}{\bar{x}\omega_n^2}\Big((-1)^n e^{\nu\bar{x}}\big(2\nu - \omega_n \bar{x}\big) - 2\nu\Big) \tag{44}$$

In the quadratic case, $f_Z(x) = x^2$ implies

$$\theta_n^Z = \chi \int_0^{\bar{x}} x^2 e^{\nu x} \sin\left(\frac{\pi n x}{\bar{x}}\right) dx$$

Apply equation (43) and evaluate at the limits:

$$\chi \int_0^{\bar{x}} x^2 e^{\nu x} \sin\left(\frac{\pi n x}{\bar{x}}\right) dx =$$
$$\chi \frac{\pi n}{\bar{x}\omega_n^3}\Big[(-1)^n e^{\nu\bar{x}}\left(-\omega_n^2\bar{x}^2 + 4\nu\omega_n\bar{x} - 2\left(3\nu^2 - \left(\frac{\pi n}{\bar{x}}\right)^2\right)\right) + 2\left(3\nu^2 - \left(\frac{\pi n}{\bar{x}}\right)^2\right)\Big]$$

□

B.4 Calculating Impulse and Cumulative Responses

Theorem 2 showed that macroeconomic aggregates are easily calculated in the frequency domain, where $\hat{Z}(s)$ is linear in the reset flow. This section demonstrates how to convert $\hat{Z}(s)$ into macroeconomic objects that are typically reported: the IRF and

CIR.

A convenient property of the frequency domain is that $\hat{Z}(s)$ can be treated as an infinite-order polynomial fraction and thus easily inverted to the time domain.[19] If its poles are simple, denote its partial fraction expansion by

$$\hat{Z}(s) = \sum_{j=0}^{\infty} \frac{\xi_j^Z}{s + \rho_j^Z} \tag{45}$$

As a convention, let the index $j = 0$ denote the zero pole, i.e., $\rho_0^Z = 0$. Therefore $\xi_0^Z = \overline{Z}$, the long-run value of the aggregate variable $Z(t)$.

Corollary 4 gives the IRF in terms of the partial fraction expansion. It also gives the CIR, which many economists use to summarize the effects of shocks in continuous-time models.

Corollary 4 (Impulse Response Function)**.** *The impulse response function* $IRF^Z(t) \equiv Z(t) - \overline{Z}$ *is*

$$Z(t) - \overline{Z} = \sum_{j=1}^{\infty} \xi_j^Z \, e^{-\rho_j^Z t}$$

for $t > 0$*, and the cumulative impulse response* CIR^Z *is*

$$CIR^Z \equiv \int_0^{\infty} \left[Z(t) - \overline{Z} \right] \, dt = \sum_{j=1}^{\infty} \frac{\xi_j^Z}{\rho_j^Z}$$

where ξ_j^Z *and* ρ_j^Z *correspond to the partial fraction expansion in equation* (45)*.*

Proof. Inverting the partial fraction expansion gives

$$Z(t) = \sum_{j=0}^{\infty} \xi_j^Z \, e^{-\rho_j^Z t}$$

Because $\rho_0^Z = 0$, the $j = 0$ term is $\xi_0^Z = \overline{Z}$. Therefore

$$Z(t) - \overline{Z} = \sum_{j=1}^{\infty} \xi_j^Z \, e^{-\rho_j^Z t}$$

[19] This is true if $f_Z(x) \in \mathcal{C}^1$. Any finite truncation of the eigenfunction sum used for computation implies that $\hat{Z}(s)$ in the computer is necessarily rational and thus has an ordinary finite partial-fraction expansion. Differentiable $f_Z(x)$ ensures that the eigenfunction weights decay sufficiently for a finite truncation to converge for $t > 0$.

Integrating the impulse response over time gives

$$CIR^Z = \int_0^\infty \left[Z(t) - \overline{Z} \right] \, dt = \sum_{j=1}^{\infty} \frac{\xi_j^Z}{\rho_j^Z}$$

□

C Closed-Form Expressions from the Stationary Distribution

This section derives the stationary distribution with drift and random resets, then uses it to calculate moments of gap adjustments and inaction times.

C.1 The Stationary Distribution

The stationary density $\bar{h}(x) \equiv \lim_{t\to\infty} h(x,t)$ and reset flow $\bar{F} \equiv \lim_{t\to\infty} F(t)$ satisfy the KFE

$$0 = \gamma\partial_x^2\bar{h}(x) - \mu\partial_x\bar{h}(x) - \eta\bar{h}(x) + \delta(x - x^*)\bar{F} \tag{46}$$

with $\bar{h}(0) = \bar{h}(\bar{x}) = 0$ and $\int_0^{\bar{x}} \bar{h}(x)\,dx = 1$.

Taking the limit of the conditional solution in equation (6) gives

$$\bar{h}(x) = \bar{F}\int_0^{\infty} G(x, x^*, t)\,dt \qquad \bar{F} = \left(\int_0^{\bar{x}}\int_0^{\infty} G(x, x^*, t)\,dt\,dx\right)^{-1} \tag{47}$$

$\bar{h}(x)$ is determined by the steady-state flow $\bar{F}$ and is independent of the initial condition. The expression for $\bar{F}$ ensures that $\bar{h}(x)$ integrates to one.

Corollary 5 gives these objects in closed form. Define

$$\nu \equiv \frac{\mu}{2\gamma} \qquad \varrho \equiv \frac{\sqrt{\mu^2 + 4\gamma\eta}}{2\gamma}$$

Corollary 5. *The stationary density is*

$$\bar{h}(x) = \frac{\bar{F}e^{\nu(x-x^*)}}{\gamma\varrho\sinh(\varrho\bar{x})}\begin{cases}\sinh(\varrho x)\sinh\left(\varrho(\bar{x} - x^*)\right) & 0 \le x \le x^* \\ \sinh(\varrho x^*)\sinh\left(\varrho(\bar{x} - x)\right) & x^* \le x \le \bar{x}\end{cases} \tag{48}$$

where

$$\bar{F} = \frac{\eta\sinh(\varrho\bar{x})}{\sinh(\varrho\bar{x}) - e^{-\nu x^*}\sinh\left(\varrho(\bar{x} - x^*)\right) - e^{\nu(\bar{x}-x^*)}\sinh(\varrho x^*)} \tag{49}$$

When $\eta = 0$ or $\varrho = 0$, the expressions in Corollary 5 are understood as their continuous limits.

Proof. Away from x^*, equation (46) is

$$\gamma \bar{h}''(x) - \mu \bar{h}'(x) - \eta \bar{h}(x) = 0$$

Substitute $\bar{h}(x) = e^{\nu x} u(x)$:

$$u''(x) - \varrho^2 u(x) = 0$$

The boundary conditions imply

$$\bar{h}(x) = \begin{cases} Ae^{\nu x} \sinh(\varrho x) & 0 \le x \le x^* \\ Be^{\nu x} \sinh\left(\varrho(\bar{x} - x)\right) & x^* \le x \le \bar{x} \end{cases}$$

for some A and B. Continuity at x^* requires

$$A \sinh(\varrho x^*) = B \sinh\left(\varrho(\bar{x} - x^*)\right)$$

Hence for some constant C,

$$\bar{h}(x) = Ce^{\nu(x-x^*)} \begin{cases} \sinh(\varrho x) \sinh\left(\varrho(\bar{x} - x^*)\right) & 0 \le x \le x^* \\ \sinh(\varrho x^*) \sinh\left(\varrho(\bar{x} - x)\right) & x^* \le x \le \bar{x} \end{cases} \tag{50}$$

Similar to the outflow at the absorbing boundaries, the probability inflow at x^* must match $\bar{F}$:

$$\gamma \left(\bar{h}'(x^{*-}) - \bar{h}'(x^{*+})\right) = \bar{F}$$

Differentiating the two components of equation (50) gives

$$\begin{aligned} \bar{h}'(x^{*+}) - \bar{h}'(x^{*-}) = & -C\varrho \left[\sinh(\varrho x^*) \cosh\left(\varrho(\bar{x} - x^*)\right)\right. \\ & \left. + \cosh(\varrho x^*) \sinh\left(\varrho(\bar{x} - x^*)\right)\right] \\ = & -C\varrho \sinh(\varrho \bar{x}) \end{aligned}$$

Therefore

$$C = \frac{\bar{F}}{\gamma \varrho \sinh(\varrho \bar{x})}$$

which proves equation (48).

The derivatives of the density at the boundaries are

$$\gamma \bar{h}'(0) = \bar{F} e^{-\nu x^*} \frac{\sinh\left(\varrho(\bar{x} - x^*)\right)}{\sinh(\varrho \bar{x})} \tag{51}$$

$$-\gamma \bar{h}'(\bar{x}) = \bar{F} e^{\nu(\bar{x} - x^*)} \frac{\sinh(\varrho x^*)}{\sinh(\varrho \bar{x})} \tag{52}$$

Integrating equation (46) over $[0, \bar{x}]$ and using $\int_0^{\bar{x}} \bar{h}(x)\, dx = 1$ gives

$$\bar{F} = \eta + \gamma \bar{h}'(0) - \gamma \bar{h}'(\bar{x})$$

Substituting equations (51) and (52) and solving for $\bar{F}$ gives equation (49). □

C.2 Adjustment and Inaction-Time Moments

This section derives in closed form a number of steady-state moments that are measured in manufacturing data by Baley and Blanco (2021).

Let τ denote the duration between two consecutive adjustments by a single agent in the steady state. Because every adjustment begins a new inaction spell and adjustments occur at rate $\bar{F}$, the average spell duration is

$$\mathbb{E}[\tau] = \frac{1}{\bar{F}} \tag{53}$$

Decomposing equation (3), the shares of adjustments occurring at the lower and upper boundaries are

$$p_0 \equiv \frac{\gamma \bar{h}'(0)}{\bar{F}} = e^{-\nu x^*} \frac{\sinh\left(\varrho(\bar{x} - x^*)\right)}{\sinh(\varrho \bar{x})} \tag{54}$$

$$p_{\bar{x}} \equiv -\frac{\gamma \bar{h}'(\bar{x})}{\bar{F}} = e^{\nu(\bar{x} - x^*)} \frac{\sinh(\varrho x^*)}{\sinh(\varrho \bar{x})} \tag{55}$$

Random adjustments account for the remaining share $\eta / \bar{F}$, so

$$\frac{\eta}{\bar{F}} + p_0 + p_{\bar{x}} = 1$$

Define the signed adjustment from state x as

$$\Delta x \equiv x^* - x$$

An adjustment from the lower boundary has size x^*, an adjustment from the upper boundary has size $x^* - \bar{x}$, and the pre-adjustment state of a randomly adjusting firm has density $\bar{h}(x)$. Define

$$M_j \equiv \int_0^{\bar{x}} (x^* - x)^j \bar{h}(x)\, dx \qquad A_j \equiv \mathbb{E}\left[(\Delta x)^j\right] \qquad M_0 = 1 \tag{56}$$

The raw adjustment moments are therefore

$$A_j = \frac{\eta}{\bar{F}} M_j + p_0 (x^*)^j + p_{\bar{x}} (x^* - \bar{x})^j \tag{57}$$

These moments can be written directly in terms of the stationary density. Let $f_j(x) \equiv (x^* - x)^j$, so $f_j(x^*) = 0$. Multiplying equation (46) by $f_j(x)$ and integrating by parts gives

$$\begin{aligned} \bar{F} A_j &= \eta M_j + \gamma \bar{h}'(0)(x^*)^j - \gamma \bar{h}'(\bar{x})(x^* - \bar{x})^j \\ &= \int_0^{\bar{x}} \left(\gamma f_j''(x) + \mu f_j'(x)\right) \bar{h}(x)\, dx \\ &= j(j-1)\gamma M_{j-2} - j\mu M_{j-1} \end{aligned} \tag{58}$$

for $j \geq 2$. For $j = 1$, the same calculation gives $\bar{F} A_1 = -\mu$. Thus, the first four raw moments are

$$A_1 = -\frac{\mu}{\bar{F}} \tag{59}$$

$$A_2 = \frac{2\gamma - 2\mu M_1}{\bar{F}} \tag{60}$$

$$A_3 = \frac{6\gamma M_1 - 3\mu M_2}{\bar{F}} \tag{61}$$

$$A_4 = \frac{12\gamma M_2 - 4\mu M_3}{\bar{F}} \tag{62}$$

In particular, mean duration and mean adjustment satisfy

$$\mathbb{E}[\Delta x] = -\mu \mathbb{E}[\tau] \tag{63}$$

To derive the joint duration–adjustment moment, define normalized duration

$$\widetilde{\tau} \equiv \frac{\tau}{\mathbb{E}[\tau]}$$

Let $\bar{a}$ denote the average elapsed time since the last adjustment among firms in the stationary cross-section. For a spell of length τ, the average elapsed time is $\tau/2$. Thus

$$\bar{a} \equiv \frac{\mathbb{E}[\tau^2]}{2\mathbb{E}[\tau]} = \frac{\mathbb{E}[\tau]}{2}\left(1 + \mathrm{CV}_\tau^2\right) \qquad \mathrm{CV}_\tau^2 \equiv \frac{\mathrm{Var}(\tau)}{\mathbb{E}[\tau]^2} \tag{64}$$

Let $x(t)$ denote the state t periods after an adjustment, and let $x^- \equiv x(\tau^-)$ denote the state immediately before the next adjustment. Between adjustments, $x(t)$ follows equation (1), beginning from $x(0) = x^*$. Applying Itô's formula to $tx(t)$ and taking expectations gives

$$\begin{aligned}
\mathbb{E}[\tau x^-] &= \mathbb{E}\left[\int_0^\tau x(t)\,dt\right] + \frac{\mu}{2}\mathbb{E}[\tau^2] \\
&= \mathbb{E}[\tau]\int_0^{\bar{x}} x\bar{h}(x)\,dx + \frac{\mu}{2}\mathbb{E}[\tau^2] \\
&= \mathbb{E}[\tau](x^* - M_1) + \frac{\mu}{2}\mathbb{E}[\tau^2]
\end{aligned} \tag{65}$$

The second equality follows because the stationary density records the share of time spent at each state, and the last uses equation (56). Since $\Delta x = x^* - x^-$, equations (64) and (65) imply

$$\mathbb{E}[\widetilde{\tau}\Delta x] = M_1 - \mu\bar{a}$$

Using $\mathbb{E}[\widetilde{\tau}] = 1$ and equation (59), the covariance between normalized duration and adjustment size is therefore

$$\mathrm{Cov}(\widetilde{\tau}, \Delta x) = M_1 + \mu\left(\mathbb{E}[\tau] - \bar{a}\right) = M_1 + \frac{\mu\mathbb{E}[\tau]}{2}\left(1 - \mathrm{CV}_\tau^2\right) \tag{66}$$

The duration moments in equations (64) and (66) can be calculated directly from the within-spell problem. Let $D_j(x) \equiv \mathbb{E}_x[\tau^j]$ and $D_0(x) \equiv 1$. For $j = 1, 2$, the

backward equations are

$$\begin{aligned} \gamma D_j''(x) + \mu D_j'(x) - \eta D_j(x) &= -j D_{j-1}(x) \\ D_j(0) = D_j(\bar{x}) &= 0 \end{aligned} \tag{67}$$

Thus $D_1(x^*) = \mathbb{E}[\tau] = 1/\bar{F}$ and $D_2(x^*) = \mathbb{E}[\tau^2]$.

The variance, standard deviation, fourth central moment, raw fourth-to-second moment ratio, and conventional central kurtosis of gap changes are

$$V_{\Delta x} \equiv A_2 - A_1^2 \tag{68}$$

$$\mathrm{SD}[\Delta x] = \sqrt{V_{\Delta x}} \tag{69}$$

$$C_{4,\Delta x} \equiv A_4 - 4A_1A_3 + 6A_1^2A_2 - 3A_1^4 \tag{70}$$

$$R_{4,\Delta x} \equiv \frac{A_4}{A_2^2} \tag{71}$$

$$\mathrm{Kur}_c[\Delta x] \equiv \frac{C_{4,\Delta x}}{V_{\Delta x}^2} \tag{72}$$

D The Investment Problem

This appendix describes the firm's dynamic problem and how to transform it from state variables (K, P) into the one-dimensional problem in normalized log capital used in the main text.[20]

D.1 The Firm's Dynamic Problem

Let $V(K, P)$ denote the firm's value. Log productivity follows $dp = g_p\,dt + \sigma_p\,dW$, so productivity in levels follows

$$dP = g_P P\,dt + \sigma_p P\,dW$$

where $g_P \equiv g_p + \frac{\sigma_p^2}{2}$. Inside the inaction region, $dK = -\zeta K\,dt$. For the extension with free adjustment opportunities at rate η (with $\eta = 0$ in the baseline), define the post-adjustment value operator by

$$\mathcal{R}V(K, P) \equiv \max_{K' \geq 0} \{V(K', P) - q(K' - K)\}$$

Firms discount at rate r, so their two-dimensional inaction-region HJB is

$$\begin{aligned} rV(K,P) = (1-\tau)P^{1-\alpha}K^{\alpha} - \zeta K V_K(K,P) + g_P P V_P(K,P) \\ + \frac{\sigma_p^2 P^2}{2} V_{PP}(K,P) + \eta\left[\mathcal{R}V(K,P) - V(K,P)\right] \end{aligned} \tag{73}$$

At either inaction boundary, value matching requires

$$V(K, P) = \mathcal{R}V(K, P) - \kappa q K^{\text{tar}}$$

Smooth pasting and optimality imply $V_K(K, P) = q$ at both boundaries and the reset point.

[20] This approach is well known; see Stokey (2008) for a textbook treatment. I include it here, using Section 6's notation, for the interested reader.

D.2 Normalization

After-tax operating profits, capital expenditure, and the fixed cost $\kappa q K^{\text{tar}}(P,\tau)$ are all homogeneous of degree one in (K,P). Therefore, the value function is as well:

$$V(cK,cP) = cV(K,P)$$

for any $c>0$. Define normalized capital and its value function by

$$\widetilde{K} \equiv \frac{K}{(1-\tau)^{1/(1-\alpha)}P} \qquad V(K,P) = (1-\tau)^{1/(1-\alpha)}P\widetilde{V}(\widetilde{K}) \tag{74}$$

This normalization removes the tax coefficient from operating profits, which become $\widetilde{K}^{\alpha}$. The fixed cost is also normalized:

$$\tilde{\kappa} \equiv \frac{\kappa q K^{\text{tar}}}{(1-\tau)^{1/(1-\alpha)}P} = \kappa q \left(\frac{\alpha}{q(r+\zeta)}\right)^{1/(1-\alpha)}$$

Thus the normalized problem is independent of τ.

The derivatives satisfy

$$V_K = \widetilde{V}' \qquad V_P = (1-\tau)^{1/(1-\alpha)}\left(\widetilde{V} - \widetilde{K}\widetilde{V}'\right) \qquad V_{PP} = \frac{(1-\tau)^{1/(1-\alpha)}}{P}\widetilde{K}^2\widetilde{V}''$$

Homogeneity also implies that the reset point is $K^*(P) = (1-\tau)^{1/(1-\alpha)}P\widetilde{K}^*$ and that

$$\frac{\mathcal{R}V(K,P)}{(1-\tau)^{1/(1-\alpha)}P} = \widetilde{V}(\widetilde{K}^*) - q(\widetilde{K}^* - \widetilde{K})$$

Substituting these expressions into equation (73) and dividing by $(1-\tau)^{1/(1-\alpha)}P$ gives the intermediate normalized HJB

$$\begin{aligned} r\widetilde{V}(\widetilde{K}) = \widetilde{K}^{\alpha} - \zeta\widetilde{K}\widetilde{V}'(\widetilde{K}) + g_P\left[\widetilde{V}(\widetilde{K}) - \widetilde{K}\widetilde{V}'(\widetilde{K})\right] \\ + \frac{\sigma_p^2}{2}\widetilde{K}^2\widetilde{V}''(\widetilde{K}) \\ + \eta\left[\widetilde{V}(\widetilde{K}^*) - q(\widetilde{K}^* - \widetilde{K}) - \widetilde{V}(\widetilde{K})\right] \end{aligned} \tag{75}$$

This step effectively transforms the discount rate. Moving $g_P\widetilde{V}$ to the left-hand side

gives

$$\begin{aligned}\rho\widetilde{V}(\widetilde{K}) &= \widetilde{K}^{\alpha} - (\zeta + g_P)\widetilde{K}\widetilde{V}'(\widetilde{K}) + \frac{\sigma_p^2}{2}\widetilde{K}^2\widetilde{V}''(\widetilde{K}) \\ &\quad + \eta\left[\widetilde{V}(\widetilde{K}^*) - q(\widetilde{K}^* - \widetilde{K}) - \widetilde{V}(\widetilde{K})\right] \\ \rho &\equiv r - g_P = r - g_p - \frac{\sigma_p^2}{2}\end{aligned} \tag{76}$$

The coefficient ρ is the original discount rate r net of the expected growth rate of productivity.

Define

$$k \equiv \log \widetilde{K} \qquad v(k) \equiv \widetilde{V}(e^k)$$

The derivatives satisfy

$$v'(k) = \widetilde{K}\widetilde{V}'(\widetilde{K}) \qquad v''(k) = \widetilde{K}\widetilde{V}'(\widetilde{K}) + \widetilde{K}^2\widetilde{V}''(\widetilde{K})$$

Using these identities in equation (76) gives

$$\begin{aligned}\rho v(k) &= e^{\alpha k} + (\mu - \sigma_p^2)v'(k) + \frac{\sigma_p^2}{2}v''(k) \\ &\quad + \eta\left[v(k^*) - q\left(e^{k^*} - e^k\right) - v(k)\right] \\ \mu &\equiv -(\zeta + g_p)\end{aligned} \tag{77}$$

The physical law of motion for log normalized capital is

$$dk = \mu\, dt - \sigma_p\, dW$$

At a costly adjustment, the firm pays $\tilde{\kappa}$ in normalized units as well as the net capital expenditure. Homogeneity therefore gives the following five conditions in normalized levels

$$\begin{aligned}\widetilde{V}(\underline{\widetilde{K}}) &= \widetilde{V}(\widetilde{K}^*) - q(\widetilde{K}^* - \underline{\widetilde{K}}) - \tilde{\kappa} \\ \widetilde{V}(\overline{\widetilde{K}}) &= \widetilde{V}(\widetilde{K}^*) - q(\widetilde{K}^* - \overline{\widetilde{K}}) - \tilde{\kappa} \\ \widetilde{V}'(\underline{\widetilde{K}}) &= \widetilde{V}'(\widetilde{K}^*) = \widetilde{V}'(\overline{\widetilde{K}}) = q\end{aligned}$$

The first two conditions impose value matching at the adjustment boundaries. The derivative conditions impose smooth pasting at both boundaries and optimality of the common reset point $\widetilde{K}^*$. Since $v'(k) = \widetilde{K}\widetilde{V}'(\widetilde{K})$, the same five conditions in logs

are

$$
\begin{aligned}
v(\underline{k}) &= v(k^*) - q\left(e^{k^*} - e^{\underline{k}}\right) - \tilde{\kappa} \\
v(\overline{k}) &= v(k^*) - q\left(e^{k^*} - e^{\overline{k}}\right) - \tilde{\kappa} \\
v'(\underline{k}) &= qe^{\underline{k}} \\
v'(k^*) &= qe^{k^*} \\
v'(\overline{k}) &= qe^{\overline{k}}
\end{aligned}
$$

E Large Corporate Tax Cuts

This appendix extends Section 6.3's tax experiment to finite tax cuts, illustrating how the abstract objects in Section 3 map to the investment model. I consider an economy in the stationary distribution $\bar{h}(x)$ and a permanent, unanticipated tax cut that shifts every firm's capital gap by $\Delta = -x^*$. Using equation (30), the new tax rate is

$$\tau_1 = 1 - (1 - \tau_0)e^{(1-\alpha)x^*}$$

Under Section 6.1's calibration, this is a cut from 26% to approximately 21.8%.

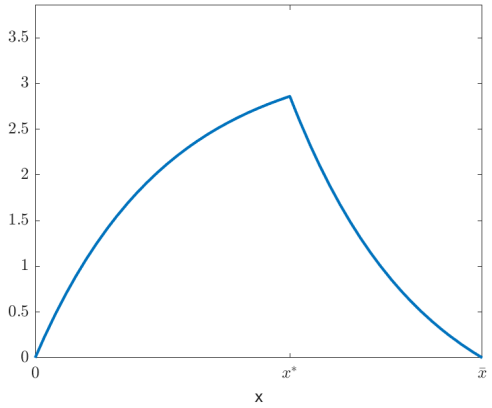
The $\Delta = -x^*$ shock pushes the $[0, x^*]$ portion of the stationary distribution (Figure 2a) to or below the lower bound at $x = 0$. These firms immediately invest in new capital and reset their capital gap to x^*. Thus, the initial condition has a point mass at x^*. The shock also shifts the remaining $[x^*, \bar{x}]$ portion of the stationary distribution to the $[0, \bar{x} - x^*]$ interval. Figure 2b plots the point-mass and density components, which together make up the initial condition $\phi(x) = h(x, 0)$.

Theorem 1 implies that the initial condition is sufficient to solve for $F(t)$, the flow of resetting firms. Figure 2c plots this function. Many firms are near the boundary at $x = 0$ immediately after the shock, so the flow remains high for a while before falling below $\bar{F}$ and then converging back to this long-run value.

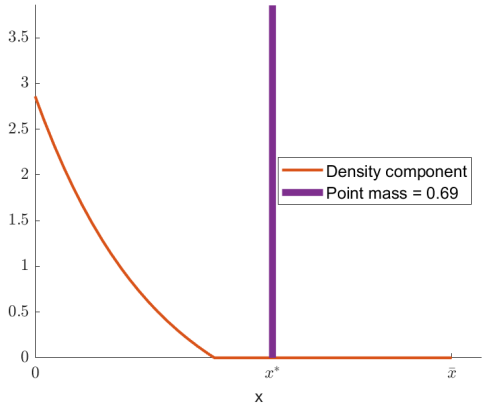
Figure 2d plots the entire distribution. Its evolution explains why the reset flow is non-monotonic. After the shock, many firms are near the boundary at $x = 0$, so the reset flow remains high. These firms quickly diffuse over the boundary and the flow falls. At the same time, the firms that reset on impact are slow to diffuse to either boundary, so the flow falls below its limiting value because most mass is far from the boundaries. In the long run, $h(x, t)$ and the reset flow converge to their stationary values.

The distribution maps the solution to data on firms' states and adjustments, but directly describes only moments such as the average state $X(t)$. Section 6.2 shows that aggregate output and capital can instead be constructed from a productivity-weighted distribution, which can also be solved using Corollary 1.

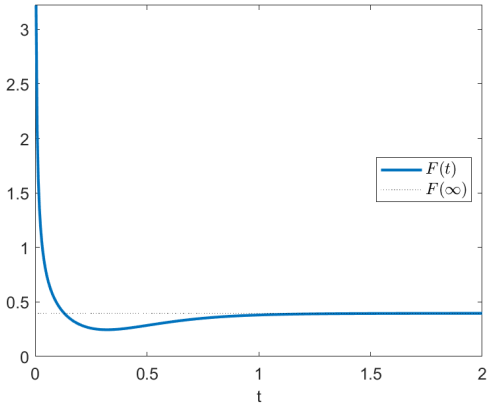
The tax experiment also provides a direct numerical validation of Theorem 3. Figure 3 compares the functional sufficient statistic with exact responses to tax cuts of 0.01, 0.10, 1, and 4 percentage points, each of which is normalized by its size for comparability. As the tax cut becomes smaller, the response converges to the theoretical



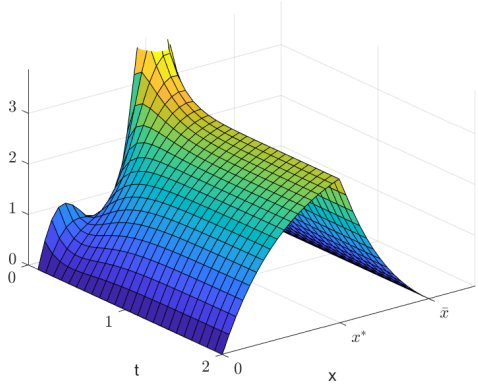
(a) Stationary distribution $h(x, \infty)$



(b) Initial condition $\phi(x)$



(c) Flow of capital-adjusting firms



(d) Dynamic distribution $h(x, t)$

Figure 2: Effects of a Finite Corporate Tax Cut

Notes: Time is measured in years. At time $t = 0$, the tax rate falls from 26% to approximately 21.8%, shifting capital gaps by $\Delta = -x^*$. The upper panels plot the stationary distribution and the post-shock initial condition. The lower panels plot the resulting reset flow and dynamic distribution for $t > 0$.

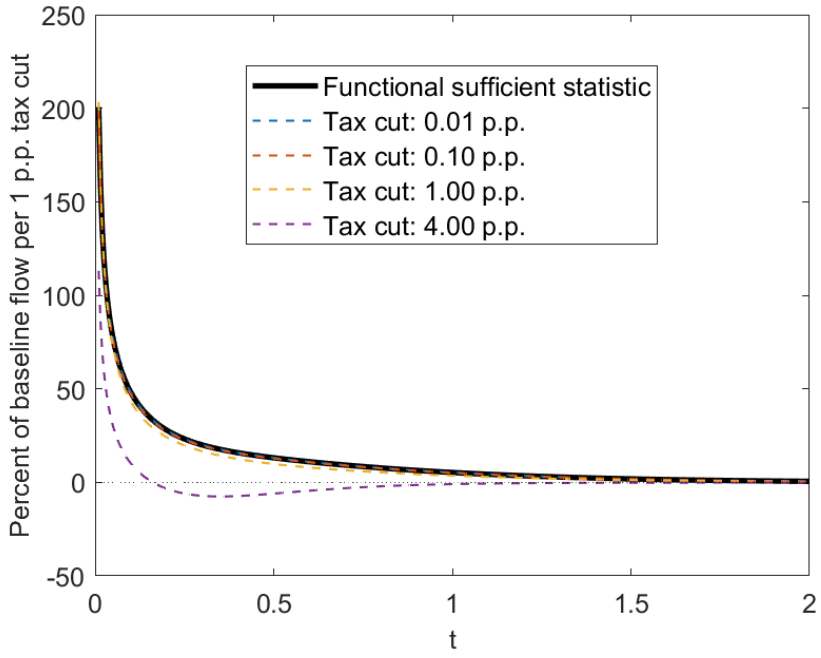


Figure 3: Tax Cuts and the Functional Sufficient Statistic

Notes: Time is measured in years. The initial tax rate is $\tau_0 = 0.26$. The solid black line is the marginal reset-flow response implied by Theorem 3, converted to a tax-cut response using equation (30). Colored dashed lines are exact finite-shock responses relative to $\bar{F}$, divided by the tax cut's size. All responses are measured as the percent change in the reset flow per one-percentage-point tax cut and plotted for $t > 0$, excluding the mass that adjusts on impact.

marginal response. The figure also reveals the limits to marginal analysis; the IRF is differently shaped for large tax cuts, where nonlinearities become quantitatively important.

F Macroeconomic Effects of Increasing the Productivity Growth Rate

This section considers a timely example: the arrival of AI, which I model as an increase in the growth rate of productivity. As discussed in Section 2.2, the general fixed cost model assumes that inaction boundaries remain fixed. Because the AI shock affects the boundaries, it is a helpful example to show that this paper's method can still be useful in such cases.

Specifically, I study a permanent, unanticipated increase in the growth rate of

aggregate productivity. Increasing g_p decreases the drift μ one-for-one, hastening the downward flow of firms' capital gaps. As a result, firms behave differently: they reset normalized capital to a higher level and increase both boundaries and the width of the inaction region.

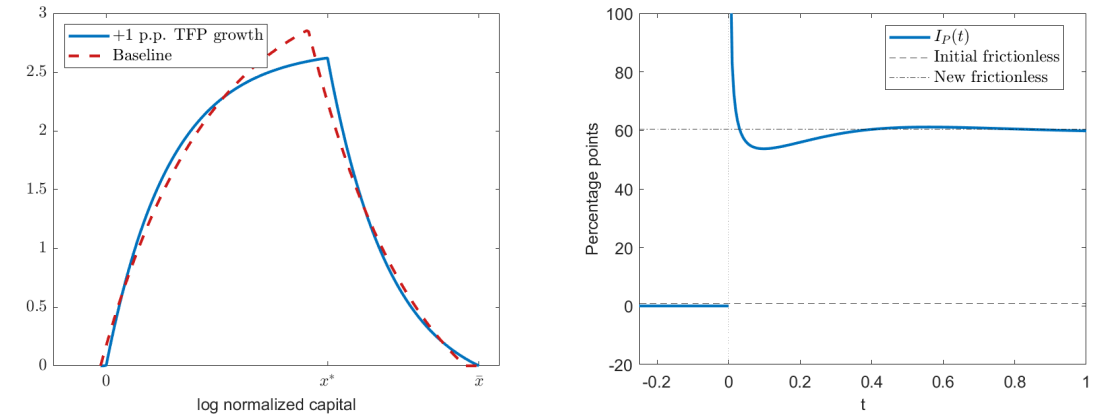
To analyze the shock using Section 3's framework, initialize the economy in the pre-shock steady state. This becomes the initial distribution, shifted relative to the post-shock inaction region $[0, \bar{x}]$. Some firms will initially be outside the post-shock inaction region; they immediately reset, creating a mass point at x^* in the initial condition $h(x, 0)$. Quantitatively, I increase annual TFP growth $(1 - \alpha)g_p$ by one percentage point. Figure 4a plots the stationary distributions before and after the shock. The change is very small, even though a one-percentage-point increase is large. The inaction boundaries and reset point barely increase, and the shape of the new stationary distribution is nearly unchanged.

Yet, even though the distribution is mostly unaffected, the shock causes substantial investment dynamics. In a frictionless economy, detrended investment $I_P(t)$ would immediately and permanently increase in response to the shock; Figure 4b plots this level shift with a dashed black line.

In contrast, if there are fixed costs of adjustment, then investment exhibits an immediate boom followed by an extended lull. Figure 4b plots this IRF in blue. The boom occurs because, on impact, the shock shifts the inaction region, raising the lower bound and causing a mass of firms to immediately invest. This boom lingers in the short run as many firms remain close to the new lower bound. The boom then decays into a lull because there is a region near the lower boundary where the initial distribution lies below the new steady state, so firms slowly diffuse into this region, partially at the expense of exiting the lower boundary. Eventually, the distribution converges to the new steady state, with aggregate investment only slightly below the frictionless level.

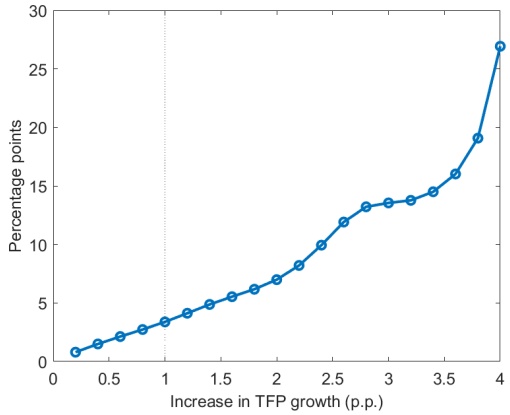
The severity of the investment lull depends on the size of the shock. When the increase in productivity growth is small, the depth of the lull is also small. But Figure 4c documents that larger increases in productivity growth produce deeper lulls. This is because the new steady state shifts more substantially from the initial distribution. However, Figure 4d documents that when the lull depth increases, the length of the lull decreases, so this spell can either be severe and short or mild and protracted.

These dynamics reveal one of the values of the analytical solution: it gives the full

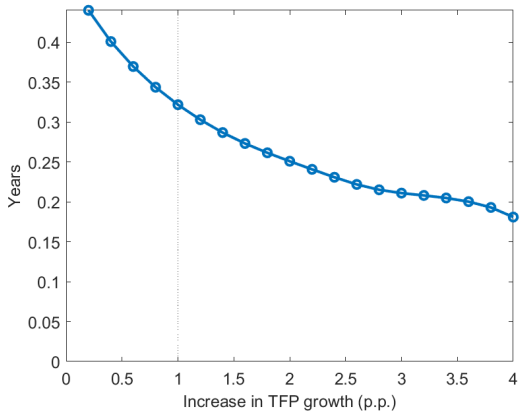


(a) Stationary Distributions

(b) Detrended Investment



(c) Lull Depth



(d) Lull Length

Figure 4: Effects of a Permanent Increase in Productivity Growth

Notes: Time is measured in years. At $t = 0$, the annual growth rate of TFP increases by one percentage point. The upper-left panel compares the stationary distributions of log normalized capital before and after the shock. The upper-right panel plots detrended investment $I_P \equiv I/\bar{P}$ in percentage points relative to its pre-shock steady state. The frictionless investment levels use $(1-\tau)\alpha K_{P,f}^{\alpha-1} = (r+\zeta)q$ and $I_{P,f} = (g_P + \zeta)K_{P,f}$. A mass of firms invest at $t = 0$, so the vertical axis is truncated. The lower panels vary the increase in annual TFP growth; dotted vertical lines mark the one-percentage-point baseline. Lull depth is the distance between the post-shock minimum of $I_P(t)$ and its new steady state, as a percent of the steady state. Lull length is the time from the shock until $I_P(t)$ first reaches its new steady state from below after the nadir.

IRF, making the boom-lull pattern clear. Moreover, it is easily calculated from the flow $F_P(t)$ without the need to solve the entire distributional dynamics.

Another value of the analytical solution is that it is easily applied to a shock of any size. Many analytical results in the literature are specific to marginal shocks (including Section 5); the full solution complements these methods by easily revealing the effects of shock size. All experiments in Figure 4 begin from the same calibrated steady state but feature different dynamic responses.

G Reinjection and the Tax Response

How much does endogenous reinjection matter? When there is zero drift: not at all for certain aggregates in symmetric models (Alvarez and Lippi, 2022). But when the state variable drifts between resets, it can substantially change the macroeconomic dynamics, which is why it is important to account for the endogenous reset flow. This appendix demonstrates this using Section 6's calibrated investment model.

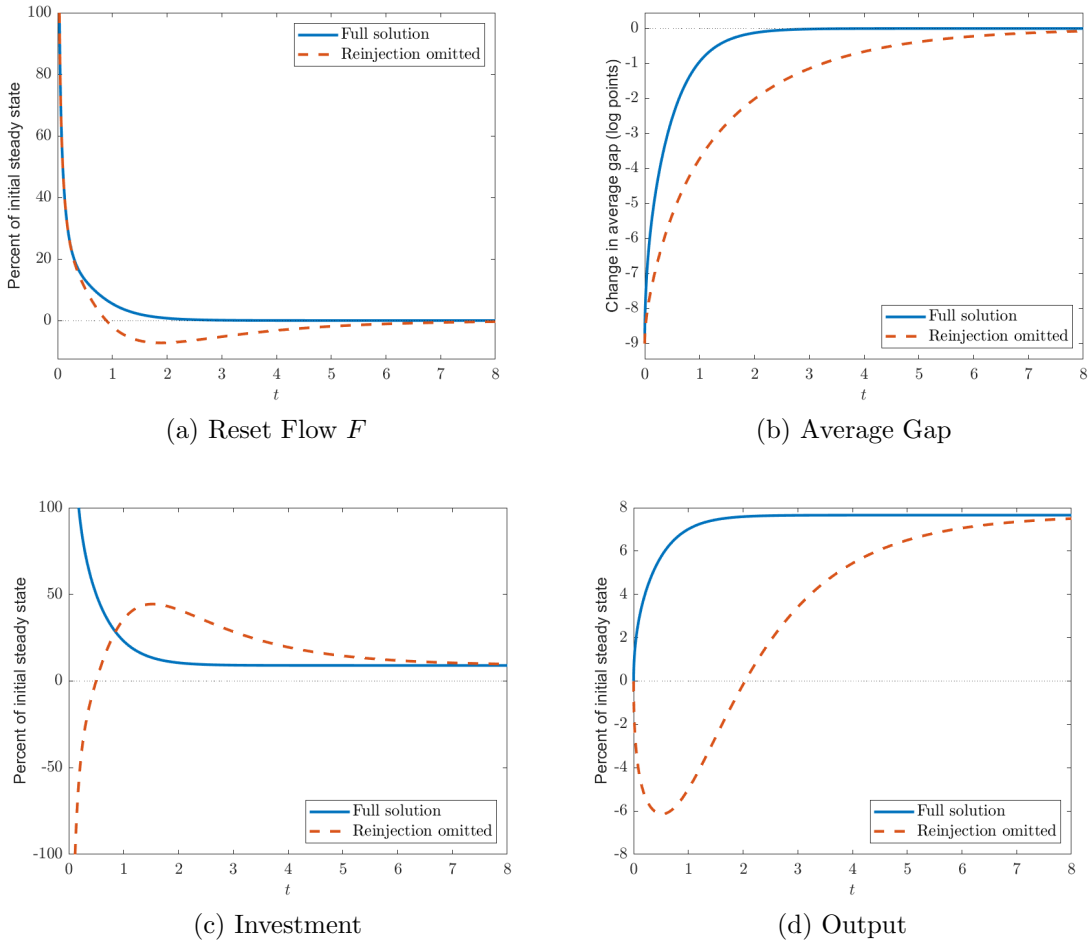
To illustrate the quantitative importance of the endogenous reset flow, I repeat Section 6.3's marginal tax-cut exercise, omitting the reinjection term from the marginal density's solution:

$$\int_0^t G(x, x^*, t-\tau) F^{\Delta}(\tau)\, d\tau$$

Otherwise, the calibration and initial condition are unchanged. Then I calculate the resulting macroeconomic dynamics, which are effectively just the behavior of agents that have never reset. Figure 5 plots these dynamics calculated without reinjection against the true responses.

Omitting reinjection creates a spurious decline in the reset flow. Figure 5a shows that the full reset-flow response remains positive as it decays, but omitting reinjected firms allows the flow to erroneously fall below the steady state. Theorem 2 implies that if the flow is calculated wrong, then an aggregate variable will be calculated wrong. Figure 5b plots the response of the average gap, which converges substantially faster in the true solution than the erroneous version that understates the reset flow.

The errors caused by omitting reinjection are further exacerbated when applied to the productivity-weighted distribution. Figure 5c shows that without reinjection, the implied investment surge is *negative* rather than positive. The tax cut causes firms to reach the lower boundary and invest. Omitting reinjection removes their existing capital from the calculated aggregate without adding their capital after adjustment, so this capital loss manifests through equation (27) as negative investment. The capital loss badly distorts the implied output IRF (Figure 5d) which features an immediate recession rather than a monotonic boom.



(a) Reset Flow F

(b) Average Gap

(c) Investment

(d) Output

Figure 5: Marginal Tax Responses with Reinjection Omitted

Notes: Responses are percentage changes from each calibration's initial steady state, scaled to a one-percentage-point tax cut. Time is measured in years. Panels (a) and (b) use the unweighted density h; panels (c) and (d) use the productivity-weighted density h_P. Solid lines give the full solution; dashed lines omit reinjection.

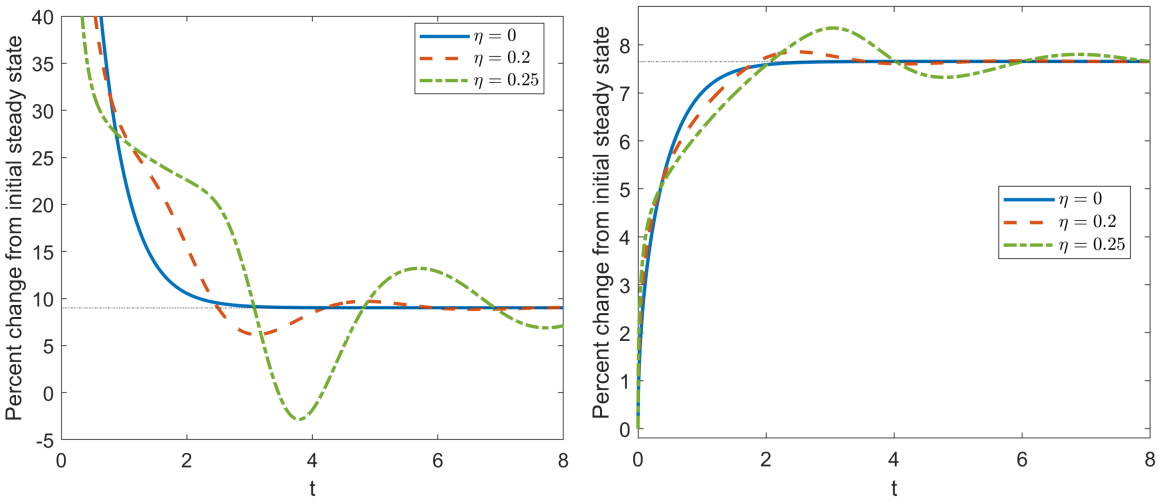
H The IRF vs. the CIR

Theorem 3 recovers the entire marginal impulse response from steady-state adjustment information. What does this add relative to existing results that characterize the CIR (Alvarez et al., 2016; Baley and Blanco, 2021)?

Models with the same CIR can have very different IRFs. To demonstrate, I consider three different parameterizations of the investment model. In each case, I fix the steady-state adjustment size and frequency, and the tax cut CIR.[21] This model has four free parameters: the free adjustment rate η, the normalized fixed cost $\tilde{\kappa}$, and the productivity process parameters g_p and σ_p. This leaves one degree of freedom, which I vary in order to demonstrate how a particular CIR is consistent with different IRFs. I summarize each of these three parameterizations with the free adjustment rate, even though multiple parameters change values. Thus $\eta = 0$ denotes the baseline.

Figure 6 demonstrates that different parameterizations can imply dissimilar IRF shapes, yet the same output CIR value. Panel 6a plots the marginal IRF of the tax shock for each of the three cases. The solid blue line corresponds to the traditional baseline case with no free adjustments; the investment response is smooth and monotonic. However, as the free adjustment rate increases, the tax cut introduces additional waves to the IRF. This is because more common free adjustments make the steady-state distribution increasingly irregular. In all cases, they converge to the same new long-run value, consistent with the common calibration targets. Panel 6b plots the corresponding output IRFs. The waves from capital accumulation appear here as well. Yet despite the different shapes, they all integrate to the same value (relative to the new steady state): the output CIR is fixed.

[21] In order to fix the CIR across alternative parameterizations, I no longer match the adjustment variance target.



(a) Investment (b) Output

Figure 6: Equal Cumulative Output Responses with Different Timing

Notes: Responses are percentage changes from each calibration's initial steady state, scaled to a one-percentage-point tax cut. Time is measured in years. Each calibration matches the baseline steady-state adjustment frequency and size, and has $CIR^{Y_P,\Delta} = 0.298$. Dotted lines denote the new steady state.